NIHR | National Institute for Health and Care Research

# Summary

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| **Reference Number** | NIHR305457 |
| **Lead Applicant** | Dr Mark Russell |
| **Research Title** | The GuideALIGN Study: A data-driven approach to monitoring the implementation of NICE guidance |
| **Plain English Summary** | **Why is this research needed?**<br>Clinical guidelines are recommendations on how to diagnose and treat medical conditions, based upon the best available evidence. However, we cannot easily check whether these recommendations are being followed in real-world care. Current methods (e.g. collecting data manually using spreadsheets) take too long and make it hard to build a national picture of care quality.<br>Importantly, we can now bring together electronic health records for the whole population of England using a system called OpenSAFELY. This does not need manual data collection, and patients cannot be identified. It gives us a unique opportunity to change how we monitor the quality of care for patients.<br><br>**What are the aims of my research?**<br>My goal is to improve patient care by increasing the use of guideline recommendations. To achieve this, I will do the following:<br>1. Create a data-driven way of monitoring guideline use in real-world care;<br>2. Use this information to make a plan that gives guideline-recommended care to patients;<br>3. Test this plan using real-world data in OpenSAFELY.<br><br>To show this, I have chosen a condition with very effective treatments, where guidelines are rarely followed. Gout is a common type of arthritis, affecting 1 in 40 adults. It causes severe pain, permanent joint damage and health problems (e.g. kidney disease) if not treated well enough. Very good medications (e.g. allopurinol) can prevent these complications, but they are not used often enough or in the right amounts.<br><br>**What will my project do?**<br>1. Check guideline use: I will use anonymous GP and hospital data for 58 million people in OpenSAFELY to see whether national treatment recommendations are being followed for people with gout. I will show how care varies throughout England; find out what determines whether patients receive good quality care; and how this affects health outcomes.<br>2. Make a plan to improve care: I will work with patients, healthcare professionals, charities and policy makers, to find areas of good care and areas for improvement. Together, we will make a plan that gives guideline-recommended care to people with gout, using digital solutions and education.<br>3. Real-world testing: I will test whether this plan improves patient care using an efficient, real-world study in OpenSAFELY. General practices (38 in total) will provide care to people with gout using either the new plan or the treatment that would usually be given. All study data will be collected automatically in OpenSAFELY, with no extra data needed from patients or doctors. The following results will be compared for patients treated for gout over a 12-month period:<br>  - Whether patients received guideline-recommended treatments (e.g. allopurinol);<br>  - If they developed complications of gout (e.g. kidney problems);<br>  - How often they visited their GP or hospital.<br><br>**How will the results be used?**<br>My research will directly benefit people with gout, by reducing pain and disability. It will help doctors and nurses give better care to their patients. It will reduce costs for the NHS and help focus resources on areas where improvement is needed most. I will be working closely with the National Institute for Health and Care Excellence (NICE) to make sure my research benefits people with other long-term conditions. I will share all of my findings openly through conferences, research papers, and by working with our professional and patient communities. |

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| | **Patient and public involvement (PPI)**<br>My research plan has been shaped by the experiences of patients, gathered through interviews, patient and carer events, and discussions with patient organisations. Throughout the project, I will be working together with patient partners, a patient and lay members group and the UK Gout Society, to create participant resources, review my findings, and support PPI strategy. |

Our purpose for collecting this data is to communicate with you about your application and have the necessary information to evaluate you for a grant. The data we collect here is collected in the public interest. Information provided here may be subject to Freedom of Information requests.

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The NIHR privacy policy includes further information including ways we may use your data, our contact details and details on your individual rights regarding how your data is used. Your data may be shared across the NIHR, including with other coordinating centres, to allow the application to be managed and for statistical analysis, and with external grant reviewers as part of the process for managing the allocation of a grant. This notice is under constant review and will be updated and / or revised based on that review as appropriate.

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## 1. Application Summary Information

| **Host organisation (which will administer any award)** |
|---|
| King's College London |

| **Partner organisation (if applicable)** | |
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| **NIHR Academy Strategic Theme** | Health data science |
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| **Partnership Fellowship** | N/A |
|---|---|

| **Advanced Fellowship application type**<br>NIHR Themed Call / Highlight Notice **OR** Career Stage |
|---|
| Recently completed or about to complete PhD |

| **Research Title** | The GuideALIGN Study: A data-driven approach to monitoring the implementation of NICE guidance |
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| **Research Type** | Clinical trial (phase 3b - effectiveness), Epidemiological |
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| **Proposed start date if grant awarded** | 01/07/2025 |
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| **Do you wish to hold an Advanced Fellowship for the full time equivalent of 2, 3, 4 or 5 years?** |
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| 5 |

| **WTE %** | 1.00 |
|---|---|

| **Duration**<br>The duration in months below for your award is based on your choice of years for the Advanced Fellowship and your chosen WTE% |
|---|
| 60 |

## 2. Applicant CV

**Degrees and Professional Qualifications**

| From | To | Qualification | Subject | Country | Institution | Class | Department |
|---|---|---|---|---|---|---|---|
| 01/2021 | 12/2023 | PhD | Medicine | United Kingdom | King's College London | Awarded with no corrections | Faculty of Life Sciences & Medicine |
| 10/2006 | 06/2012 | MBBChir | Medicine | United Kingdom | University of Cambridge | Distinction | School of Clinical Medicine |
| 10/2006 | 06/2009 | MA(Hons) | Pre-clinical Medical Sciences | United Kingdom | University of Cambridge | 1st Class | Trinity College |

**Present and previous positions**
(including from/to dates, job title and employer)

| From | To | Position | Organisation | WTE % Research |
|---|---|---|---|---|
| 07/2024 | - | NIHR Clinical Lecturer in Rheumatology | King's College London | 50 |
| 01/2024 | 06/2024 | Postdoctoral Clinical Research Fellow | King's College London | 100 |
| 01/2021 | 12/2023 | NIHR Doctoral Fellow | King's College London | 100 |
| 10/2019 | 12/2020 | Rheumatology Specialist Trainee Year 5/6 | King's College Hospital NHS Foundation Trust | 0 |
| 09/2018 | 10/2019 | Rheumatology Specialist Trainee Year 4 | Lewisham and Greenwich NHS Trust | 0 |
| 08/2017 | 09/2018 | Rheumatology Specialist Trainee Year 3 and NIHR Academic Clinical Fellow | Guy's and St Thomas' NHS Foundation Trust | 25 |
| 08/2015 | 08/2017 | NIHR Academic Clinical Fellow in Rheumatology and Core Medical Trainee | St George's University Hospitals NHS Foundation Trust | 25 |
| 09/2014 | 07/2015 | Hospital Medical Officer Year 3 | Monash Medical Centre, Melbourne | 0 |
| 08/2013 | 08/2014 | Foundation Year 2 Trainee | East and North Herts NHS Trust | 0 |
| 08/2012 | 08/2013 | Foundation Year 1 Trainee | University College London Hospitals NHS Foundation Trust | 0 |

**Research Grants held**

| Title of Grant | Role in the Research Grant | Total Grant £ | Source of Grant (Organisation Name) | Duration (In Months) | Start Date of Funding | End Date of Funding |
|---|---|---|---|---|---|---|
| TalLOR - Trial of Patient-Initiated Care Leading to Improved Outcomes in Rheumatology | I am costed into an NIHR HSDR grant, led by Professor Laura Coates at the University of | 2312351 | NIHR Health and Social Care Delivery Research | 49 | 01/04/2024 | 01/05/2028 |

| Title of Grant | Role in the Research Grant | Total Grant £ | Source of Grant (Organisation Name) | Duration (In Months) | Start Date of Funding | End Date of Funding |
|---|---|---|---|---|---|---|
| | Oxford. I am utilising electronic health data, via OpenSAFELY, to evaluate the effectiveness of the NHS England Patient-Initiated Follow-Up (PIFU) programme for rheumatologic al conditions in England. This study is ongoing. | | | | | |
| Removing Barriers to Remission in the Biosimilar Era | I was co-lead investigator and lead analyst on an industry-funded, competitive grant, exploring variation in biologic prescribing for rheumatoid arthritis. This provided me with 6 months of dedicated post-doctoral research. I designed the project, conducted all analyses (using National Audit data), and have submitted outputs for publication (*The Lancet Rheumatology*). | 79884 | Sandoz UK | 6 | 01/01/2024 | 30/06/2024 |
| The PREVENT Study: Preventing hospital | I was principal investigator and lead applicant on a NIHR Doctoral | 432822 | NIHR | 36 | 01/01/2021 | 31/12/2023 |

| Title of Grant | Role in the Research Grant | Total Grant £ | Source of Grant (Organisation Name) | Duration (In Months) | Start Date of Funding | End Date of Funding |
|---|---|---|---|---|---|---|
| admissions attributable to gout | Fellowship. I used big data analyses and implementation science methodology to improve the quality of care for people hospitalised for gout. I designed the project, led all analyses, and published all outputs, including:<br>• https://doi.org/10.1016/j.lanepe.2022.100416<br>• https://doi.org/10.1016/S2665-9913(23)00206-0<br>• https://doi.org/10.1093/rheumatology/kead574<br>• https://doi.org/10.1093/rheumatology/keac638<br>• https://doi.org/10.3899/jrheum.211203<br>• https://doi.org/10.1093/rheumatology/keab539 | | | | | |

| Title of Grant | Role in the Research Grant | Total Grant £ | Source of Grant (Organisation Name) | Duration (In Months) | Start Date of Funding | End Date of Funding |
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**Publication Record**

(include the list of authors, article title, journal name, and volume)
This information can be edited via the "My Research Outputs" page of your ARAMIS account.

| Publications |
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| Nagra, D., Bechman, K., Russell, MD., Yang, Z., Adas, MA., Molbanti, H., Khan, A., Wincup, C., Alveyn, E., Baldwin, C., Rutherford, AI., Subesinghe, S., Cope, A. and Galloway, JB.. (2024) 'Pneumococcal vaccine uptake in patients with inflammatory arthritis: a single centre cohort study.', *Rheumatology (Oxford, England)*. |
| Yang, Z., Arumalla, N., Alveyn, E., Gallagher, S., Price, E., Russell, MD., Bechman, K., Norton, S. and Galloway, J.. (2024) 'Trust performance in managing inflammatory arthritis over time in England and Wales: a latent class analysis approach.', *Rheumatology advances in practice*, 8(2). |
| Brown, AD., Fisher, L., Curtis, HJ., Wiedemann, M., Hulme, WJ., Speed, V., Hopcroft, LEM., Cunningham, C., Costello, RE., Galloway, JB., Russell, MD., Bechman, K., Kurt, Z., Croker, R., Wood, C., Walker, AJ., Schaffer, AL., Bacon, SCJ., Mehrkar, A., Hickman, G., Bates, C., Cockburn, J., Parry, J., Hester, F., Harper, S., Goldacre, B. and MacKenna, B.. (2024) 'OpenSAFELY: The impact of COVID-19 on azathioprine, leflunomide and methotrexate monitoring, and factors associated with change in monitoring rate.', *British journal of clinical pharmacology*. |
| Russell, MD., Yang, Z., Walter, B., Alveyn, E., Bechman, K., Miracle, A., Nagra, D., Adas, MA., Norton, S., Cope, AP., Langan, SM. and Galloway, JB.. (2024) 'The influence of safety warnings on the prescribing of JAK inhibitors.', *The Lancet. Rheumatology*, 6(3). |
| Russell, MD., Dey, M., Flint, J., Davie, P., Allen, A., Crossley, A., Frishman, M., Gayed, M., Hodson, K., Khamashta, M., Moore, L., Panchal, S., Piper, M., Reid, C., Saxby, K., Schreiber, K., Senvar, N., Tosounidou, S., van de Venne, M., Warburton, L., Williams, D., Yee, CS., Gordon, C. and Giles, I.. (2024) 'British Society of Rheumatology guideline working group response to European Medicines Agency safety update on Hydroxychloroquine.', *Rheumatology (Oxford, England)*, 63(2). |
| Nagra, D., Bechman, K., Russell, MD., Yang, Z., Adas, M., Subesinghe, S., Rutherford, A., Alveyn, E., Patel, S., Wincup, C., Mahto, A., Baldwin, C., Karafotias, I., Cope, A., Norton, S. and Galloway, J.. (2024) 'No Waning of Pneumococcal Vaccine Responses over Time in People with Inflammatory Arthritis: Findings from a Single Centre Cohort.', *Vaccines*, 12(1). |
| Bechman, K., Yang, Z., Adas, M., Nagra, D., Soykan Uguzlar, A., Russell, MD., Wilson, N., Steer, S., Norton, S. and Galloway, J.. (2023) 'The incidence of uveitis in patients with axial spondylarthritis treated with biologics or targeted synthetics: a systematic review and network meta-analysis.', *Arthritis & rheumatology (Hoboken, N.J.)*. |
| Lim, KK., Ng, K., Balachandran, S., Russell, MD., Boalch, A., Sinclair, A., Coker, B., Vinnakota, K., Mansoor, R., Douiri, A., Marks, LV. and Sacks, S.. (2023) 'Measuring the impact of monoclonal antibody therapies.', *Frontiers in medicine*, 10. |
| Nagra, D., Bechman, K., Adas, M., Yang, Z., Alveyn, E., Subesinghe, S., Rutherford, A., Allen, V., Patel, S., Russell, MD., Cope, A., Norton, S. and Galloway, J.. (2023) 'A Systematic Review and Meta-Analysis of Anti-Rheumatic Drugs and Pneumococcal Vaccine Immunogenicity in Inflammatory Arthritis.', *Vaccines*, 11(11). |
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| Bechman, K., Cook, E., Alveyn, E., Houssien, A., Stevens, M., Russell, MD., Adas, M., Amlani-Hatcher, P., Norton, S., Lempp, H., Ledingham, J., Galloway, J. and Walker-Bone, K.. (2023) 'Occupational impacts of early inflammatory arthritis: results from the National Early Inflammatory Arthritis Audit.', *Rheumatology (Oxford, England)*. |
| Russell, MD., Massey, J., Roddy, E., MacKenna, B., Bacon, S., Goldacre, B., Andrews, CD., Hickman, G., Mehrkar, A., Mahto, A., Rutherford, AI., Patel, S., Adas, MA., Alveyn, E., Nagra, D., Bechman, K., Ledingham, JM., Hudson, J., Norton, S., Cope, AP. and Galloway, JB.. (2023) 'Gout incidence and management during the COVID- |

**Publication Record**

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| Publications |
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| 19 pandemic in England, UK: a nationwide observational study using OpenSAFELY.’, *The Lancet. Rheumatology*, 5(10). |
| Bechman, K., Russell, MD. and Galloway, JB.. (2023) ‘Predicting COVID-19 vaccination response in populations who are immunosuppressed.’, *The Lancet. Rheumatology*, 5(8). |
| Nagra, D., Russell, MD., Alveyn, E., Birring, SS., Elias, D., Balachandran, S. and Galloway, JB.. (2023) ‘Diagnosing Camurati-Engelmann disease-the age of whole-exome sequencing.’, *Rheumatology (Oxford, England)*, 62(7). |
| Russell, MD., Stovin, C., Alveyn, E., Adeyemi, O., Chan, CKD., Patel, V., Adas, MA., Atzeni, F., Ng, KKH., Rutherford, AI., Norton, S., Cope, AP. and Galloway, JB.. (2023) ‘JAK inhibitors and the risk of malignancy: a meta-analysis across disease indications.’, *Annals of the rheumatic diseases*. |
| Schreiber, K., Frishman, M., Russell, MD., Dey, M., Flint, J., Allen, A., Crossley, A., Gayed, M., Hodson, K., Khamashta, M., Moore, L., Panchal, S., Piper, M., Reid, C., Saxby, K., Senvar, N., Tosounidou, S., van de Venne, M., Warburton, L., Williams, D., Yee, CS., Gordon, C. and Giles, I.. (2023) ‘British Society for Rheumatology guideline on prescribing drugs in pregnancy and breastfeeding: comorbidity medications used in rheumatology practice.’, *Rheumatology (Oxford, England)*, 62(4). |
| Russell, MD., Dey, M., Flint, J., Davie, P., Allen, A., Crossley, A., Frishman, M., Gayed, M., Hodson, K., Khamashta, M., Moore, L., Panchal, S., Piper, M., Reid, C., Saxby, K., Schreiber, K., Senvar, N., Tosounidou, S., van de Venne, M., Warburton, L., Williams, D., Yee, CS., Gordon, C. and Giles, I.. (2023) ‘British Society for Rheumatology guideline on prescribing drugs in pregnancy and breastfeeding: immunomodulatory anti-rheumatic drugs and corticosteroids.’, *Rheumatology (Oxford, England)*, 62(4). |
| Pates, K., Periselneris, J., Russell, MD., Mehra, V., Schelenz, S. and Galloway, JB.. (2023) ‘Rising incidence of Pneumocystis pneumonia: A population-level descriptive ecological study in England.’, *The Journal of infection*, 86(4). |
| Adas, MA., Russell, MD., Cook, E., Alveyn, E., Hannah, J., Balachandran, S., Oyebanjo, S., Amlani-Hatcher, P., Ledingham, J., Norton, S. and Galloway, JB.. (2023) ‘COVID-19 admissions and mortality in patients with early inflammatory arthritis: Results from a UK national cohort.’, *Rheumatology (Oxford, England)*. |
| Mukherji, P., Adas, MA., Clarke, B., Galloway, JB., Mulvey, T., Norton, S., Turner, J., Russell, MD., Lempp, H. and Li, S.. (2022) ‘Changing trends in ethnicity and academic performance: observational cohort data from a UK medical school.’, *BMJ open*, 12(12). |
| Adas, MA., Norton, S., Balachandran, S., Alveyn, E., Russell, MD., Esterine, T., Amlani-Hatcher, P., Oyebanjo, S., Lempp, H., Ledingham, J., Kumar, K., Galloway, JB. and Dubey, S.. (2022) ‘Worse outcomes linked to ethnicity for early inflammatory arthritis in England and Wales: a national cohort study.’, *Rheumatology (Oxford, England)*, 62(1). |
| Russell, MD., Roddy, E., Rutherford, AI., Ellis, B., Norton, S., Douiri, A., Gulliford, MC., Cope, AP. and Galloway, JB.. (2022) ‘Treat-to-target urate-lowering therapy and hospitalisations for gout: results from a nationwide cohort study in England.’, *Rheumatology (Oxford, England)*. |
| Russell, MD., Galloway, JB., Andrews, CD., MacKenna, B., Goldacre, B., Mehrkar, A., Curtis, HJ., Butler-Cole, B., O'Dwyer, T., Qureshi, S., Ledingham, JM., Mahto, A., Rutherford, AI., Adas, MA., Alveyn, E., Norton, S., Cope, AP. and Bechman, K.. (2022) ‘Incidence and management of inflammatory arthritis in England before and during the COVID-19 pandemic: a population-level cohort study using OpenSAFELY.’, *The Lancet. Rheumatology*, 4(12). |
| Russell, MD., Rutherford, AI., Ellis, B., Norton, S., Douiri, A., Gulliford, MC., Cope, AP. and Galloway, JB.. (2022) ‘Management of gout following 2016/2017 European (EULAR) and British (BSR) guidelines: An interrupted time-series analysis in the United Kingdom.’, *The Lancet regional health. Europe*, 18. |
| Russell, MD., Nagra, D., Clarke, BD., Balachandran, S., Buazon, A., Boalch, A., Bechman, K., Adas, MA., Alveyn, EG., Rutherford, AI. and Galloway, JB.. (2022) ‘Hospitalizations for Acute Gout: Process Mapping the Inpatient Journey and Identifying Predictors of Admission.’, *The Journal of rheumatology*, 49(7). |
| Russell, MD., Coath, F., Yates, M., Bechman, K., Norton, S., Galloway, JB., Ledingham, J., Sengupta, R. and Gaffney, K.. (2022) ‘Diagnostic delay is common for patients with axial spondyloarthritis: results from the National Early Inflammatory Arthritis Audit.’, *Rheumatology (Oxford, England)*, 61(2). |

**Publication Record**

(include the list of authors, article title, journal name, and volume)
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| Publications |
|---|
| Russell, MD., Clarke, BD., Roddy, E. and Galloway, JB.. (2021) 'Improving outcomes for patients hospitalized with gout: a systematic review.', *Rheumatology (Oxford, England)*, 61(1). |
| Adas, MA., Allen, VB., Yates, M., Bechman, K., Clarke, BD., Russell, MD., Rutherford, AI., Cope, AP., Norton, S. and Galloway, JB.. (2021) 'A systematic review and network meta-analysis on the safety of early interventional treatments in rheumatoid arthritis.', *Rheumatology (Oxford, England)*. |
| Nagra, D., Russell, MD., Rosmini, S., Sado, D., Buazon, A., Shafi, T., Hamlyn, E., Sandhu, G., Rutherford, AI. and Galloway, JB.. (2021) 'A Kawasaki-like illness in an adult with recent SARS-CoV-2 infection.', *Rheumatology advances in practice*, 5(2). |
| Russell, MD., Bukhari, M., Shah, AM. and Galloway, JB.. (2021) 'Colchicine and the heart: old friends, old foes.', *Rheumatology (Oxford, England)*, 60(5). |
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| Galloway, JB., Norton, S., Barker, RD., Brookes, A., Carey, I., Clarke, BD., Jina, R., Reid, C., Russell, MD., Sneep, R., Sugarman, L., Williams, S., Yates, M., Teo, J., Shah, AM. and Cantle, F.. (2020) 'A clinical risk score to identify patients with COVID-19 at high risk of critical care admission or death: An observational cohort study.', *The Journal of infection*, 81(2). |
| Russell, MD., Yates, M., Bechman, K., Rutherford, AI., Subesinghe, S., Lanyon, P. and Galloway, JB.. (2019) 'Rising Incidence of Acute Hospital Admissions due to Gout.', *The Journal of rheumatology*, 47(4). |
| Russell, MD., Bukhari, M. and Galloway, J.. (2019) 'The price of good health care.', *Rheumatology (Oxford, England)*, 58(6). |
| Russell, MD., Barrick, TR., Howe, FA. and Sofat, N.. (2018) 'Reduced anterior cingulate grey matter volume in painful hand osteoarthritis.', *Rheumatology international*, 38(8). |
| Sofat, N., Harrison, A., Russell, MD., Ayis, S., Kiely, PD., Baker, EH., Barrick, TR. and Howe, FA.. (2017) 'The effect of pregabalin or duloxetine on arthritis pain: a clinical and mechanistic study in people with hand osteoarthritis.', *Journal of pain research*, 10. |
| Russell, MD., Singh-Ranger, D. and Crosbie, J.. (2014) 'An AIDS-defining neoplasm in a 30-year-old man on highly active antiretroviral therapy: a forgotten diagnosis?', *BMJ Case Reports*, 2014. |
| Russell, MD., Young, AM. and Karri, SK.. (2013) 'Biomarkers of pediatric brain tumors.', *Frontiers in pediatrics*, 1. |

**Relevant Prizes, Awards and other Academic Distinctions**

Please note the information for additional prizes or awards is automatically populated from information provided in the "Update CV" section of the "Manage My Details" page, which you can update by following this link https://aramis.nihr.ac.uk/MyAccount/UpdateCV.aspx

| Date (mm/yyyy) | Source | Prize | Role of Applicant |
|---|---|---|---|
| 01/04/2024 | British Society for Rheumatology | Shortlisted for Best Practice Award | Lead researcher |
| 01/10/2022 | King's College London | Postgraduate Research Excellence Award | Lead researcher and award recipient |
| 01/06/2022 | Royal Society of Medicine | Eric Bywaters Prize Winner | Lead researcher and award recipient |
| 01/05/2018 | British Society for Rheumatology | Young Investigator Award Winner | Lead researcher and award recipient |
| 01/08/2013 | North Thames Foundation School | North Thames Foundation Trainee Merit Award | Award recipient |

**Relevant Prizes, Awards and other Academic Distinctions**

Please note the information for additional prizes or awards is automatically populated from information provided in the "Update CV" section of the "Manage My Details" page, which you can update by following this link https://aramis.nihr.ac.uk/MyAccount/UpdateCV.aspx

| Date (mm/yyyy) | Source | Prize | Role of Applicant |
|---|---|---|---|
| 01/06/2009 | Trinity College, University of Cambridge | Tripos Prize in Medicine (2007, 2008, 2009) | Award recipient |
| 01/06/2009 | Trinity College, University of Cambridge | Rashbass Prize in Physiology (2008 and 2009) | Award recipient |
| 01/06/2009 | Trinity College, University of Cambridge | Nominated for a Research Scholarship at Trinity College | Award recipient |
| 01/06/2008 | Trinity College, University of Cambridge | Awarded a Senior Scholarship at Trinity College | Award recipient |
| 01/06/2007 | Trinity College, University of Cambridge | Awarded a Junior Scholarship at Trinity College | Award recipient |

**ORCID iD** 0000-0001-8171-7772

## 3. Applicant Research Background

| **Professional Background** | Medically Qualified |
|---|---|

| **Specialty (if Medically Qualified)** |
|---|
| Rheumatology |

| **Do you have a GMC number?** | Yes |
|---|---|

| **GMC number**<br>Please supply a valid GMC number | 7281098 |
|---|---|

**Please describe your research career to date**

I am a rheumatologist and NIHR Clinical Lecturer with a strong track record in health data analysis and quality improvement.

**Key research experience:**

I was awarded an NIHR Doctoral Fellowship at King's College London (KCL) in 2020. I combined big data analysis (OpenSAFELY; CPRD; Hospital Episode Statistics) and implementation science methodology, to improve the quality of care for people hospitalised for gout. During my Fellowship, I published 28 peer-reviewed publications, including 4 first-author publications in *The Lancet* specialty journals. I was awarded prizes nationally (Royal Society of Medicine, British Society for Rheumatology) and regionally (King's Research Excellence Award). I was interviewed by the *BBC*, *The Times* and *The Guardian*, which helped raise the profile of gout as a sub-optimally managed condition.

Currently, I am an NIHR Clinical Lecturer at KCL, leading a project with the Healthcare Quality Improvement Partnership (HQIP) to transform National Clinical Audit programmes in England and Wales using big data (Russell *et al. Lancet Rheumatology.* https://doi.org/10.1016/S2665-9913(22)00305-8). I am co-investigator on an NIHR HSDR grant (£2.3 million), led by Professor Laura Coates (University of Oxford), in which I am evaluating the real-world effectiveness of the NHS England Patient-Initiated Follow-Up programme using electronic health record data (OpenSAFELY; Hospital Episode Statistics). This project includes a pragmatic, multi-centre RCT (30 sites), which provides me with an excellent grounding in the skillsets I will be using in this Advanced Fellowship.

In January 2024, I was awarded a competitive, industry-funded grant to fund 6 months of post-doctoral research. I used National Clinical Audit data to demonstrate that under-served groups (particularly ethnic minorities and older adults) are much less likely to receive biologic medications for rheumatoid arthritis than other patients (under peer-review, *Lancet Rheumatology*).

**Teaching and supervisory experience:**

I thoroughly enjoy teaching, and regularly deliver clinically-focused seminars and lectures on Quality Improvement for the MBBS programme at KCL. I have lectured for the Integrated Academic Training programme on using health data to improve patient outcomes. I examine for the MBBS programme at KCL, and lead a project to develop written examination questions using Large-Language Models. I provide research supervision to an NIHR Pre-Doctoral Fellow, methodological support to 3 PhD students undertaking health data analyses, and I will be taking on a formal PhD supervisory role within the next 12 months. During my Doctoral Fellowship, I delivered over 50 educational talks to general practitioners throughout the UK on how to optimise gout management.

**Leadership and management experience:**

I developed many of the attributes required of an effective leader and manager when designing and implementing a management pathway for people hospitalised for gout flares. I engaged with patients, clinicians and managers to address site-specific barriers and facilitators of optimal care. I worked with Trust executives and the Integrated Care Board to ensure the pathway was adequately resourced and sustainable. I supervised a team of multi-

disciplinary clinicians to deliver effective, holistic patient care. The service I developed was shortlisted for a *Best Practice Award* by the British Society for Rheumatology, and has been implemented at three other hospitals nationally (Guy's and St Thomas' Hospital, Princess Royal University Hospital, Stockport NHS Foundation Trust).

During my Doctoral Fellowship, I was given the opportunity to lead the evidence review for a national guideline on prescribing immunomodulatory drugs in pregnancy, commissioned by the British Society for Rheumatology. I synthesised evidence from over 27,000 articles, managed a team of 4 evidence reviewers, and provided methodological oversight. I liaised with a working group of 24 multi-disciplinary experts and patient representatives. I published the guideline as first author (most-cited manuscript in Rheumatology, 2023: *https://doi.org/10.1093/rheumatology/keac551*), and I am regularly in correspondence about the implications of the guideline. I was invited by the British Society for Rheumatology to support rheumatology trainees in taking on guideline roles, and I have been invited to join the working group of another national guideline.

**Patient and public engagement:**

Patient involvement has been fundamental to the success of my doctoral and post-doctoral research - the goal of which is to improve the quality of care for people with arthritis and other long-term conditions. I work closely with patient organisations (including as a Trustee of the UK Gout Society) to ensure my research benefits patients and the wider public. I have been interviewed about my research on BBC Radio 4 (*Inside Health* – James Gallagher), recorded podcasts for the British Society for Rheumatology, Facebook Live events for the National Axial Spondyloarthritis Society, and speaker events for Versus Arthritis. The experiences of patients – gathered through interviews, stakeholder workshops, PPIE events and direct clinical care – have shaped all aspects of this application.

**Other skills:**

I have been trained in epidemiology and statistical methodology (London School of Hygiene and Tropical Medicine; University of Aberdeen; NIHR BRC), implementation science (University of Oxford; KCL), and the safe use of health data (Office for National Statistics). I am actively involved in the NIHR Clinical Trials Facility at King's College Hospital, and have been a co-investigator on 8 studies (including multi-centre RCTs). I was awarded the British Society for Rheumatology *Young Investigator Award* in 2018.

**Impact of an Advanced Fellowship on my career:**

I am committed to improving the quality of care for people with chronic rheumatic diseases. An NIHR Advanced Fellowship will provide me with the best possible opportunity of achieving this goal and of becoming a leader in data-driven quality improvement. It will help to secure me a tenured position at King's College London, and allow me to dedicate 80% of my time to research. Building on my combined experience in big data analysis, quality improvement and clinical rheumatology, I would be at the forefront of using electronic health data to optimise patient care. My research is highly relevant to organisations including NICE, HQIP and NHS England. The expertise I develop during this Fellowship will enable me to establish myself as a leading academic collaborator in this field, supported by a training programme that spans health data science, complex intervention development and pragmatic trial design. No other Fellowship programme provides me this level of opportunity.

| **Title of research degree** | The PREVENT Study: Preventing hospital admissions attributable to gout |
|---|---|

| **Has the degree already been awarded?** | Yes |
|---|---|

| **Date of Award** | 01/01/2024 |
|---|---|

| **Has this application, or a similar application, previously been submitted to this or any other funding body?** |
|---|
| No |

**Contextual Factors**
Please use this question to detail any contextual factors you wish to make the Selection Committee aware of. NIHR wants to know about any circumstances so that they may take them into consideration during the

assessment of your application. Contextual factors may include:

- Career breaks due to parental leave, or periods of illness.
- Reduced time spent undertaking research due to a disability or caring responsibilities. This could include any physical or mental difficulty that may have impacted your research career. These are situations that have a significant impact on your ability to undertake research.
- Reduced opportunities to career support e.g. mentorship, and limited opportunities to undertake prior research and training.
- Impact of the COVID-19 pandemic on your research career.

Please also use this section to detail any other factors that may have impacted your research career not listed in the examples provided. The impact on your career to date will be specific to your particular circumstances but could include such impacts as limited opportunities to obtain grant funding, or fewer publications. In general terms, mitigating factors should be significant, and relevant.

NIHR acknowledges that you may be reluctant, or uncomfortable disclosing relevant information that is sensitive. However, you should bear in mind that we are unable to take into account factors that you do not disclose. Please be assured that information provided by you is sensitive and will be treated confidentially and in line with General Data and Protection Regulations (GDPR).

## 4. Plain English Summary of Research

**Why is this research needed?**

Clinical guidelines are recommendations on how to diagnose and treat medical conditions, based upon the best available evidence. However, we cannot easily check whether these recommendations are being followed in real-world care. Current methods (e.g. collecting data manually using spreadsheets) take too long and make it hard to build a national picture of care quality.

Importantly, we can now bring together electronic health records for the whole population of England using a system called OpenSAFELY. This does not need manual data collection, and patients cannot be identified. It gives us a unique opportunity to change how we monitor the quality of care for patients.

**What are the aims of my research?**

My goal is to improve patient care by increasing the use of guideline recommendations. To achieve this, I will do the following:

1. Create a data-driven way of monitoring guideline use in real-world care;
2. Use this information to make a plan that gives guideline-recommended care to patients;
3. Test this plan using real-world data in OpenSAFELY.

To show this, I have chosen a condition with very effective treatments, where guidelines are rarely followed. Gout is a common type of arthritis, affecting 1 in 40 adults. It causes severe pain, permanent joint damage and health problems (e.g. kidney disease) if not treated well enough. Very good medications (e.g. allopurinol) can prevent these complications, but they are not used often enough or in the right amounts.

**What will my project do?**

1. Check guideline use: I will use anonymous GP and hospital data for 58 million people in OpenSAFELY to see whether national treatment recommendations are being followed for people with gout. I will show how care varies throughout England; find out what determines whether patients receive good quality care; and how this affects health outcomes.
2. Make a plan to improve care: I will work with patients, healthcare professionals, charities and policy makers, to find areas of good care and areas for improvement. Together, we will make a plan that gives guideline-recommended care to people with gout, using digital solutions and education.
3. Real-world testing: I will test whether this plan improves patient care using an efficient, real-world study in OpenSAFELY. General practices (38 in total) will provide care to people with gout using either the new plan or the treatment that would usually be given. All study data will be collected automatically in OpenSAFELY, with no extra data needed from patients or doctors. The following results will be compared for patients treated for gout over a 12-month period:
    - Whether patients received guideline-recommended treatments (e.g. allopurinol);
    - If they developed complications of gout (e.g. kidney problems);
    - How often they visited their GP or hospital.

**How will the results be used?**

My research will directly benefit people with gout, by reducing pain and disability. It will help doctors and nurses give better care to their patients. It will reduce costs for the NHS and help focus resources on areas where improvement is needed most. I will be working closely with the National Institute for Health and Care Excellence (NICE) to make sure my research benefits people with other long-term conditions. I will share all of my findings openly through conferences, research papers, and by working with our professional and patient communities.

**Patient and public involvement (PPI)**

My research plan has been shaped by the experiences of patients, gathered through interviews, patient and carer

events, and discussions with patient organisations. Throughout the project, I will be working together with patient partners, a patient and lay members group and the UK Gout Society, to create participant resources, review my findings, and support PPI strategy.

## 5. Scientific Abstract


**Background**

The National Institute for Health and Care Excellence (NICE) has published over 300 clinical guidelines, containing evidence-based recommendations for best practice care. However, the ability to monitor real-world uptake of these recommendations and demonstrate impact for patients is limited. Existing approaches rely upon manual data collection, which is time-consuming and provides only a snapshot of performance. This makes it challenging to detect variation in care and target interventions towards reducing healthcare inequality.

Many of these limitations could be overcome by using electronic health data that are captured routinely in clinical practice. Through the OpenSAFELY Trusted Research Environment, we now have the ability to analyse health data in near real-time for 58 million people, covering more than 99% of England's population. This creates the potential for continuous evaluation of care quality across multiple disease areas.

**Aims and objectives**

I will develop a data-driven approach to monitoring the uptake of NICE guidance, demonstrating real-world impact for patients, and facilitating efficient evaluation of quality improvement interventions. To develop this methodology, I have chosen gout as an exemplar disease with a persistent evidence-practice gap.

I will complete three workstreams:

1. Develop a reproducible analytic framework to monitor the implementation of NICE guidance;
2. Design a multi-faceted intervention to maximise uptake of guideline-recommended care;
3. Evaluate this intervention using a data-enabled, cluster-randomised trial in OpenSAFELY.

**Methods *(timelines for delivery)***

Workstream 1 *(months 0-12)*: Using anonymised GP and hospital data for 58 million people in OpenSAFELY, I will systematically map recommendations from the NICE Gout Guideline and analyse temporal trends in their implementation. I will use regression-modelling approaches to identify predictors of sub-optimal care, and evaluate the impact on outcomes including comorbidities and hospitalisations. Data-driven dashboards will feedback performance against NICE standards at national, regional and local levels.

Workstream 2 *(months 12-24)*: I will convene focus groups with broad stakeholder representation to identify barriers and facilitators of optimal gout care. I will collaborate with implementation science experts at King's College London to develop an evidence-based intervention package that provides NICE-recommended care for people with gout, incorporating digital solutions within electronic health record software and educational content.

Workstream 3 *(months 24-60)*: I will conduct a data-enabled, cluster-randomised trial in OpenSAFELY to evaluate the intervention from Workstream 2. General practices (n=38) will be randomised 1:1 to receive this intervention or usual care. All study data will be automatically captured in OpenSAFELY and analysed using the framework developed in Workstream 1. Outcomes will include:

- The uptake of NICE-recommended gout treatments (e.g. allopurinol);
- The incidence of gout-associated comorbidities (e.g. kidney disease);
- Consultation/hospitalisation rates.

**Anticipated impact**

My research will directly benefit patients with gout, by reducing pain and disability. Better disease control will prevent avoidable hospitalisations and complications. For policy makers, it will help target strategies towards reducing healthcare inequality. I will be collaborating with NICE throughout my Fellowship to ensure this methodology is readily adaptable for other guidelines and diseases. I will share all of my analysis code openly, and communicate my findings through conferences, research papers, social media, and PPIE events.

## 6. Detailed Research Plan

### 1. Problem

Clinical guidelines summarise evidence-based recommendations on best practice care. However, these recommendations are often slow to be adopted in practice,[1,2,26] and **we are limited in our ability to monitor the quality of care provided to patients**. Existing mechanisms rely upon manual clinical audit,[3] which is time-consuming and biased towards departments with greater resources.[4] Local audits are sporadically performed and rarely published, while national audit programmes are lacking for many conditions.

**We face similar challenges when evaluating strategies designed to improve the quality of care**. Trials are costly and take years to produce results. Outcome data are usually manually collected, which is burdensome for participants and researchers. Moreover, trial settings may not adequately reflect real-world practice.

My Fellowship seeks to address these limitations by harnessing electronic health record data that are captured routinely during clinical care. **I will generate the methodology to evaluate care quality in near real-time without any need for manual data collection.** This will enable us to quantify the impact of guideline-recommended care on patient outcomes, and analyse factors associated with sub-optimal care. I will demonstrate how a data-enabled trial can evaluate quality improvement interventions efficiently using entirely routinely-collected data. My proposal aligns closely with NIHR's vision to harness data-enabled trials,[5] and NICE's strategy for data-driven guideline evaluation.[6]

To achieve these aims, I will use the **OpenSAFELY Trusted Research Environment**. Through OpenSAFELY, we can now analyse electronic health data in near real-time for **58 million people**, covering >99% of England's population. Privacy safeguards are greater than for any other data platform: no individual patient data are visible to researchers, and all analysis code and platform activity are publicly logged.

To develop this methodology, **I have chosen gout as an exemplar disease with a persistent evidence-practice gap**. Gout is a common inflammatory disease affecting 1 in 40 adults, characterised by severe joint pain, disability and early mortality.[7] It is a hallmark of the metabolic syndrome, with patients often experiencing multiple long-term conditions, including chronic kidney and cardiovascular disease.[8,9] There are highly-effective, urate-lowering therapies (ULT, e.g. allopurinol) that can cure symptoms and prevent complications. However, in my NIHR Doctoral research, I showed that only one in three people with gout in the UK are prescribed these treatments.[10] The National Institute for Health and Care Excellence (NICE) recently published its first guideline for the management of gout, with the aim of improving care.[11] Current limitations mean that we do not know whether these recommendations are being followed or if they have improved patient outcomes.

**Using this guideline as proof-of-principle**, I will develop a data-driven framework that enables us to:

i. Monitor real-world uptake of guideline recommendations;
ii. Quantify the impact for patients;
iii. Evaluate interventions designed to deliver best practice care.

### 2. Importance

I am **collaborating with NICE** to ensure the methodology I develop is adaptable across multiple guidelines and diseases. My proposal will enable continuous evaluation of care quality, serving as an exemplar for other healthcare systems worldwide. This capability is essential, not only to NICE and policy makers, but to the 28 National Clinical Audit programmes in England and Wales that rely heavily upon manual data. We know that data gaps exist for under-served groups and under-resourced departments.[4] By utilising routinely-captured data, we can quality assure care more comprehensively and target interventions towards **reducing healthcare inequality**.

By developing this methodology for a disease with persistently sub-standard care, **patients will directly benefit from my research**. Of 1.7 million people with gout in the UK, only a third are prescribed guideline-recommended treatments.[10,12] I will develop an intervention that delivers best practice care to patients, resulting in less pain, fewer complications, and improved quality-of-life.

For healthcare professionals, the intervention will reduce unnecessary workload and **enable them to achieve better outcomes for their patients**. Electronic dashboards will provide feedback on performance against NICE standards – a powerful motivator for change.[13] Better disease control will prevent avoidable hospitalisations and complications, producing **cost-savings for the NHS**.[14]

The data-enabled trial design I am utilising will provide researchers with an efficient mechanism through which complex interventions can be evaluated in real-world care. **I will share my analysis code and methodology openly** to support efficient re-use.

**3. Existing evidence**

Existing approaches for monitoring the real-world uptake of guideline recommendations rely heavily upon manual clinical audit,[3] which is labour-intensive and provides only a snapshot of performance.[15,16] National Audit programmes can help to build a national picture of care quality, but with several limitations:

- No data are captured on patients not enrolled in these audits;[1]
- Missing data varies markedly by healthcare provider, and associates with poor care quality;[4]
- National Audit programmes are absent for many conditions (e.g. gout).

**Electronic health record data can overcome many of these limitations.** Clinically-coded information on diagnoses, investigations and prescriptions can be linked across multiple data sources and analysed to evaluate care quality. Existing data platforms (e.g. CPRD) have been available for over 20 years; however, their suitability for evaluating guideline implementation is limited: data are often outdated (e.g. >2 years for hospital data) and are present for only a minority of the population.[18,19]

The introduction of the OpenSAFELY Trusted Research Environment during the COVID-19 pandemic has transformed our ability to deliver health data analyses for the whole of England:

- There is data coverage for >99% of England's population;[20]
- Primary care data are updated weekly, and secondary care data monthly;[20]
- There is linkage to numerous additional datasets, including high-cost drugs and referral-to-treatment times;[20]
- Privacy safeguards are greater than for any other platform.[20,53]

**In my pilot work, I showed that OpenSAFELY can be used to evaluate the impact of the pandemic on long-term conditions**.[1,2] I demonstrated that inflammatory arthritis diagnoses fell by 40% during the early pandemic, while highlighting the remarkable adaptation of the NHS in the face of unprecedented pressures.[1,2] These analyses required no manual data collection; the number of captured diagnoses was 5-times greater than the existing (manually-collected) National Audit programme;[22] and these analyses can be updated regularly using electronic dashboards linked to OpenSAFELY.

**NHS England recently expanded OpenSAFELY's use beyond COVID-19-related projects**,[23] making this proposal timely and relevant. I will use this incredible resource to develop a data-driven approach to evaluating care quality in near real-time. I will develop this methodology using the recently published NICE guideline for the management of gout.[11] **By demonstrating this approach works for gout, the methodology will be applicable to many other guidelines and conditions.**

Gout is unique in that there are low-cost (£20 annually per patient), well-tolerated, urate-lowering medications (ULT, e.g. allopurinol) that prevent symptoms, disability and complications.[17,24] Guidelines recommend that ULT should be offered to all people with gout,[8,11] but care has remained woefully sub-standard. I demonstrated that hospitalisations for gout have doubled in England since 2006, despite many of these admissions being preventable with guideline-recommended treatments.[14,25] In my Doctoral Fellowship, I showed that only 34% of people with gout in the UK were prescribed ULT in 2022,[2,10] while only 36% achieved the blood urate levels necessary to prevent complications.[10] NICE guidance for gout management was introduced in June 2022 with the aim of improving care, but it remains unclear whether these recommendations are being followed. The methodology I develop will enable us to evaluate this in near real-time, highlighting variation between regions and patient groups, and quantifying the impact on health outcomes.

What is also clear from existing evidence is that **guidelines need to be accompanied by effective implementation strategies**.[10,26] I showed that guidelines from specialist societies (BSR, EULAR) had no meaningful impact on care for people with gout in the UK.[10] A recent systematic review highlighted that nurse-led education approaches can improve gout care in randomised-controlled trial settings, assuming sufficient resources are available.[17,21] However, this is unlikely to be scalable within the resource envelope of the NHS – a challenge I encountered when trying to obtain NHS funding to sustain the nurse-led, hospital gout management pathway I developed in my Doctoral Fellowship.

Some of the most effective strategies for implementing complex healthcare interventions are digital solutions that streamline care and reduce unnecessary workload.[27] Evidence from small studies outside the UK have shown that electronic health record solutions (e.g. decision-support tools and electronic prompts) can improve outcomes for people with gout.[28-30] Despite their potential benefit and scalability, **digital interventions to optimise gout management have not been evaluated at scale in the NHS**. I will address this knowledge gap using an efficient, data-enabled trial design.

Traditional approaches to evaluating these interventions would be costly and resource-intensive. In 2021, NIHR outlined the potential for routinely-collected data to increase the speed, efficiency and generalisability of trials.[31] In a recent proof-of-concept trial, a digital intervention designed to reduce the spread of COVID-19 was evaluated using entirely routinely-collected data in OpenSAFELY.[32] **I will demonstrate how a similar, data-enabled study design can evaluate quality improvement interventions in real-world care.**

**4. Aims and objectives**

My hypothesis is that real-world data in the OpenSAFELY Trusted Research Environment can be used to monitor the implementation of NICE guidance, investigate drivers of healthcare inequality, and evaluate quality improvement interventions efficiently.

I will complete three workstreams, using gout as an exemplar disease to address the following objectives:

1. **Monitor the implementation of NICE guidance in near real-time**, by mapping recommendations and outcomes using health data for >99% of England's population.

2. **Develop a multi-faceted intervention to maximise the uptake of NICE guidance***,* incorporating digital solutions and education.

3. **Evaluate the implementation and effectiveness of this intervention** using an efficient, data-enabled, cluster-randomised trial in OpenSAFELY.

**5. Project plan**

**Workstream 1 (*months 0-12*): Monitoring NICE guidance implementation**

**Aim**

To develop a data-driven approach to monitoring real-world uptake of NICE guidance, highlighting variation in care, and evaluating impact for patients.

I will develop this methodology using the NICE guideline for the management of gout. I am collaborating with NICE to ensure the methodology is readily adaptable for other guidelines and diseases.

**Methods**

Data sources: I will use anonymised electronic health record data from primary and secondary care for >99% of England's population, linked securely via OpenSAFELY. I will publish my analysis code (Python/Stata/Markdown) via GitHub to enable efficient re-use.

Study design: I will use an observational cohort design to evaluate the uptake and impact of NICE recommendations for all people with recorded gout diagnoses in England.

To achieve this, I will:

1. Develop a variables library, by curating NHS data into simple, re-usable variables that make it easy to identify someone with gout and associated treatments.
2. Systematically map recommendations from the NICE gout guideline,[11] including:
    i. Initiation of ULT;
    ii. Serum urate monitoring;
    iii. Attainment of urate targets.
3. Describe temporal trends in the proportion of patients who receive NICE-recommended gout care,

nationally and separated by:
   i. Region;
   ii. Age/sex;
   iii. Ethnicity;
   iv. Index of multiple deprivation quintile;
   v. Comorbidity burden.
4. Use interrupted time-series and change-point detection methodology,[33] to evaluate whether metrics improved following the publication of NICE guidance.
5. Use regression-modelling approaches, including penalized regression (e.g. LASSO), to investigate predictors of sub-optimal care, including characteristics associated with under-served groups:
   i. Ethnic minority groups;
   ii. Socioeconomically-disadvantaged groups;
   iii. People with multiple long-term conditions.
6. Use multivariable Cox and Poisson regression to analyse whether guideline-recommended care associates with improved outcomes, including:
   i. Reductions in flares;
   ii. Fewer consultations/hospitalisations;
   iii. Reduced incidence/progression of gout-associated comorbidities, including kidney and cardiovascular disease.
7. Generate publicly-available dashboards via *OpenSAFELY Reports*, to feedback performance against NICE standards to clinicians, policy makers and patients.
8. Share this methodology with my collaborators at NICE, and adapt it for other guidelines.
9. Outline how a similar approach might work for other UK nations (e.g. via the SAIL Databank in Wales).

Sample size: I will include all people with incident or prevalent gout diagnoses. From my pilot analyses in OpenSAFELY, the point prevalence and incidence of gout in 2023 were 3.2% and 1.44 per 1,000 adults, respectively;[2] equating to 1.45 million prevalent diagnoses and 65,120 annual incident diagnoses available for analysis. This will provide 96% power to detect a 1% change in ULT initiation in the year pre- vs post-NICE guideline implementation. Descriptive data will be reported with 95% confidence intervals for subgroups, including by age, sex and ethnicity. For smaller sub-groups (e.g. women aged <60 years with gout), descriptive data will be reported when the number of individuals is >400, which provides binomial exact confidence intervals with maximum width +/- 5%.

**Risk mitigation**

Data access: I have already been granted approval by NHS England to analyse gout management using OpenSAFELY. OpenSAFELY's remit now covers non-COVID-19-related projects,[23] and I have requested approval to cover all analyses, with the support of my collaborators at OpenSAFELY.

Privacy: No individual patient data will be visible or transferred to me.[20] Analysis code will be developed using dummy data. Only summary-level outputs will be released following strict disclosivity checks. All platform activity will be publicly-logged for transparency.

Code development: From my pilot work, I have developed analysis code/codelists for many of the variables I require, for which there are low levels of missingness.

**Personal development**

To further my skills in computational analysis of complex health datasets, I will complete a part-time MSc in Health Data Science (LSHTM) and a training placement with the OpenSAFELY team (University of Oxford).

**Workstream 2 (*months 12-24*): Developing an intervention**

**Aim**

To develop a multi-faceted intervention that provides NICE-recommended care for people with gout. I will use a person-based approach to develop this intervention,[34,35] following the principles outlined in the MRC/NIHR complex intervention development framework.[36] The intervention will be deliverable as part of routine care by primary care clinicians, where the vast majority of gout is managed.

**Methods**

Stakeholders: I will establish a core stakeholder group, who I will consult throughout the stages listed below,

including people with gout, healthcare professionals, commissioners and third-sector representatives.

Focus groups: Three focus groups will be convened to encourage open discussion among those with shared experiences and develop a shared understanding of the challenges of achieving optimal gout care:

- Focus group one will include people with gout, carers and relatives. I will use the links I have formed with patient organisations (including the UK Gout Society) to purposively sample patients from diverse backgrounds, including ethnic minority and under-served communities, older patients, and those with multiple long-term conditions.
- Focus group two will include healthcare professionals from a broad range of localities and disciplines, including GPs, rheumatologists, practice nurses, primary care-based pharmacists and physician associates.
- Focus group three will include representatives from Integrated Care Systems, primary care software providers and patient organisations, to explore how to embed interventions within existing NHS systems.

Focus groups will be held in-person, with 8-10 participants each, and facilitated by a trained patient and qualitative methodologist. I will develop a topic guide with my patient advisory group, and use this to explore stakeholders' experiences, including:

1. Barriers and facilitators of implementing treat-to-target ULT in primary care, particularly for under-served groups;
2. Optimal interventions and implementation strategies to address identified barriers;
3. Whether these interventions are deliverable/scalable within existing NHS resources.

In-line with a person-based approach to intervention development, a set of guiding principles will be formulated with stakeholders to summarise the ways in which the intervention can maximise engagement and support context-specific behaviour change among healthcare professionals and patients.[34,35] Logic models will be developed, with underlying behaviours mapped to the COM-B framework;[37,38] highlighting the inputs, activities and mechanisms of change required to optimise outcomes (e.g. ULT initiation) and deliver impact for patients (e.g. fewer gout flares).[39]

Intervention development: Through my collaborations with health psychologists and implementation science experts (Centre for Implementation Science, KCL / Implementation Support Team, NICE), intervention components with an existing evidence base will be adapted for use in the context of optimising gout management.[21] Contextual factors will be assessed using the Consolidated Framework for Implementation Research (CFIR), with the CFIR-ERIC matching tool used to select appropriate implementation strategies.[40]

While the finalised intervention components will be selected during the development process, from my pilot work with stakeholders I anticipate that the core components will include:

- **Electronic health record solutions** that make it easier for primary care clinicians to deliver NICE-recommended gout care. This will utilise existing functionality within the software used in primary care in England (SystmOne, EMIS, Ardens), and will not require new software development:
  - NICE-aligned consultation templates and decision-support tools;
  - Electronic prompts to management guidance and patient education material;
  - Patient-recall reminders, to streamline the titration of ULT.
- **Educational content** for patients, empowering them to self-manage their gout:
  - Videos on gout and its management, which I will develop with the UK Gout Society and British Society for Rheumatology;
  - Signposting to patient information leaflets (Versus Arthritis) and e-Health diaries (e.g. Gout Central).
- **Electronic dashboards** that provide feedback on performance against NICE-aligned quality metrics:
  - National/regional benchmarking, using the OpenSAFELY dashboards from Workstream 1;
  - Practice-level reports via SystmOne/EMIS/Ardens.

**Care champions** will be identified within practices to support intervention uptake among colleagues. Practices will be encouraged to utilise **allied health professionals** (practice nurses, PCN pharmacists, physician associates), where available, in the monitoring of urate levels and titration-to-target.[17,21]

Intervention piloting/refinement: Intervention components will be piloted with GPs, allied health professionals and patients, using think-aloud interviews to generate a detailed understanding of user engagement.[41] Potential barriers to successful implementation will be identified, and intervention components modified to optimise their utility and acceptability.

**Risk mitigation**

Development time**:** All digital intervention components will utilise functionality already available within primary care software (SystmOne/EMIS/Ardens). This will greatly reduce development time and ensure familiarity amongst clinicians. I have begun scoping work to develop these solutions (including NICE-aligned decision-support tools with Ardens).

Stakeholder buy-in**:** I have built an extensive network of collaborators, including patients, primary/secondary care clinicians, allied health professionals, and organisations including NICE, NHS England and Ardens, to maximise input from a broad range of stakeholders.

**Personal development**

I will complete the Developing and Evaluating Complex Interventions course at the National Centre for Research Methods, led by the authors of the MRC/NIHR Complex Intervention Development Framework.

**Workstream 3 (*months 24-60*): Evaluating the intervention**

**Aim**

To conduct an efficient, data-enabled, cluster-randomised trial in a primary care setting, to evaluate the implementation and effectiveness of the intervention from Workstream 2. All study data will be automatically captured in OpenSAFELY and analysed using the code I develop in Workstream 1. No manual data collection will be required.

This methodology will be readily adaptable for other interventions and diseases, and will be shared openly including with my collaborators at NICE.

**Methods**

Study design: Two-arm, parallel-group, cluster-randomised trial, with general practices as the unit of randomisation. I have selected this study design as the intervention will be delivered at practice-level, reducing treatment contamination between individuals.

Target population: People aged ≥18 years, registered with general practices in England, who consult a primary care clinician for gout. Every patient consulting for gout at participating practices will be eligible; there will be no exclusion criteria.

Intervention: Participating practices will be randomised 1:1 to receive:

- The intervention package from Workstream 2; or
- Usual care.

The intervention will be delivered by primary care clinicians as part of routine care. It will serve as a guide to delivering NICE-recommended gout care; however, all management decisions will be at the discretion of the primary care team.

The intervention will be implemented remotely within participating practices: an initial training session will outline how to use the intervention components to deliver NICE-recommended gout care; a follow-up session will be held after two weeks, to consolidate training.

Randomisation/masking: Practices will be allocated by minimisation (conducted via the KCL Clinical Trials Unit), stratified by region and practice list-size.[42] Healthcare professionals will not be blinded to trial-arm allocation, nor will study members who implement the intervention. After implementation, pseudonymised identifiers of randomised practices will be linked securely to other data sources in OpenSAFELY. After linkage, the study team will not be able to identify practices and will be blinded to outcomes.

Outcomes aligned to NICE recommendations for gout management,[11] capturable using routinely-collected data sources in OpenSAFELY, have been chosen:

*Primary outcome:*

- Proportion of people consulting for gout who newly initiate ULT (allopurinol, febuxostat, benzbromarone, probenecid) within 6 months.

*Secondary outcomes:*

- Proportion of people who are adherent to ULT (proportion of days covered by ULT prescriptions ≥0.8)[12];
- Proportion of people prescribed concomitant flare prophylaxis during ULT initiation/titration;[11]
- Proportion of people who attain serum urate levels ≤360 micromol/L within 6 months;[11]
- Rate of urate level monitoring;
- Rate of primary care consultations for gout flares;
- Rate of hospitalisations for gout flares;
- Proportion of people consulting for gout who develop incident myocardial infarction, stroke, CKD stages 3-5, or death due to these causes.

*Safety outcomes:*

- Proportion of people who develop incident Stevens-Johnson syndrome, toxic epidermal necrolysis, erythema multiforme or DRESS.[8]

Justification for primary outcome: ULT initiation is an important metric of whether NICE guidance has been implemented, which is the primary objective of my intervention. It is reliably captured in primary care data, and extensive evidence demonstrates that early ULT initiation improves patient outcomes.[8,17,43-45] I considered other primary outcomes with my collaborators. Clinical outcomes (e.g. gout flares) can take 1-2 years to improve after ULT initiation, which would be less feasible as a primary outcome within the study timeframe.[17,46] Urate target attainment is also essential for optimal disease control;[17] however, from my pilot work, 45% of patients did not have urate levels performed within 6 months of ULT initiation,[2] which could bias outcomes if this differed between study arms. I have retained urate target attainment and flare rate as secondary outcomes, and I will explore how to handle missing urate data using methods including multiple imputation.

Statistical analysis: Analyses will follow an intention-to-treat principle, and will include all individuals within participating practices who consult for gout within 12 months of practice randomisation. Independence estimating equations will be used to estimate treatment effects for primary and secondary outcomes, with weighting to account for informative cluster size.[47,48] Analyses will be adjusted for participant and cluster-level covariates, including practice region and list-size. Sensitivity analyses will include estimations of cluster-average treatment effects using unweighted cluster-level analysis, and mixed-effects models with general practice as a random intercept.[48]

Sample size*:* From my pilot data in OpenSAFELY, there was a yearly average of 29 consultations for gout per general practice in England between 2018-2024. The proportion of patients who newly-initiated ULT within 6 months was 40% (ICC 0.077). The recruitment of 38 practices (19 practices per arm) would provide 92% power (alpha 0.05) to detect a 20-percentage-point increase in ULT initiation in the intervention group.

Feasibility statement: I sought extensive feedback from my collaborators and the NIHR RSS to inform my choice of a data-enabled, cluster-randomised trial:

- *Transferrable*: This approach will serve as an efficient, adaptable method for evaluating other quality improvement interventions across different diseases. It will provide real-world data on health outcomes without any need for manual data collection.
- *Precedent*: Previous data-enabled, cluster-randomised trials in OpenSAFELY and CPRD have shown very efficient recruitment and data collection, relative to traditional approaches.[32,49]
- *Recruitment:* I have allocated up to 12 months to recruit 38 practices. A previous data-enabled, cluster-randomised trial in CPRD, led by one of my collaborators (Gulliford *et al.* BMJ), recruited 80 general practices within 9 months.[49] I have engaged the support of the primary care lead for the South London Research Delivery Network, who will help coordinate recruitment, including throughout other regions of England. I have an extensive network of contacts in primary care from which to invite practices to participate, having delivered educational talks on gout management to over 2,000 general practitioners throughout the UK.
- *Pragmatic*: My intervention will implement NICE-recommended care for a condition that is common but poorly managed at baseline, using software (SystmOne, EMIS, Ardens) and treatments (e.g. allopurinol) that primary care clinicians are familiar with.

Internal pilot: To mitigate against unexpectedly slow recruitment, I will conduct an internal pilot after recruiting 10 practices. A red/amber/green system will be used to determine whether to progress to the full study:

- Green: recruiting practices at 100% of target rate – continue recruitment;

- Amber: recruiting practices at 60-100% – explore mechanisms of increasing recruitment;
- Red: recruiting practices at <60% – stop recruitment, and proceed with pilot analyses.

If criteria for progression are not met, further recruitment will stop, and outcomes will be analysed for recruited practices. These pilot data will inform future applications for funding, and enable me to modify my intervention and assess methods of increasing practice engagement.

Nested qualitative implementation study: I have costed in the support of a post-doctoral research assistant with qualitative expertise to conduct a nested qualitative study. This will provide more granular insight into implementation outcomes, including users' engagement with intervention components and modifications necessary to support wider implementation. Patients and healthcare professionals will be purposively recruited from participating practices with broad geographic and sociodemographic range, and invited to participate in semi-structured interviews. Normalisation Process Theory will inform interview topic guides,[50] written with my patient advisory group. Approximately 10-15 patients and 10-15 healthcare professionals will be interviewed to allow data saturation.[51] Thematic analysis will be conducted on these data.[52]

**Personal development**

I will complete a training course on cluster-randomised trials (LSHTM), and undertake a research visit to the University of Tsukuba, Japan, to learn from clinical epidemiologists and quality improvement experts in a different healthcare system.

**6. Dissemination, outputs and impact**

Healthcare outputs**:** My analysis code and methodology will be shared openly, including with my collaborators at NICE. They will serve as a transparent, adaptable framework through which:

1. Care quality can be monitored in near real-time across multiple disease areas;
2. Patient impact can be quantified - particularly for under-served groups, for whom data gaps exist currently;
3. Strategies to deliver best practice care can be evaluated efficiently.

I will be working directly with NICE to focus implementation-support activities towards reducing healthcare inequality.

Academic outputs: I will publish my findings from each workstream in high-impact, peer-reviewed journals with wide audiences. I will present my findings at national and international conferences, including to primary care and quality improvement audiences.

Patient outputs: By adopting a data-driven, implementation science approach, it will maximise the likelihood of my outputs entering our health system and benefitting patients. I will use multiple approaches to inform study participants, patients and the wider population, utilising collaborations with patient organisations (UK Gout Society, Versus Arthritis), professional organisations (British Society for Rheumatology, Royal College of General Practitioners) and public bodies (NICE, HQIP):

1. Plain language summaries to accompany all publications, co-written with my patient advisory group;
2. Social media summaries;
3. Public engagement events, via NIHR, RCGP and BSR networks;
4. Research open-days and science festivals;
5. Newsletters.

Impact statement**:** The goal of my Fellowship is to improve patient care by maximising the uptake of guideline-recommended care:

- My research will **directly benefit patients** with gout, through reductions in pain, disability and complications (e.g. erosive joint damage and kidney disease).
- For **NHS clinicians**, the intervention I develop will reduce unnecessary workload and serve as a framework for delivering best practice care for other diseases.
- For **commissioners**, better disease control will reduce the number of avoidable consultations, hospitalisations and complications.
- For **quality assurers** (e.g. CQC, HQIP), it will provide the information needed to highlight unwarranted variation in care.
- For **policy makers** (e.g. NICE, NHS England), it will provide quantifiable evidence of patient impact, and

enable resources to be focused on areas where improvement is needed most.
- For **researchers**, my data-enabled trial design will serve as a pragmatic, efficient method for evaluating interventions in real-world care.

**7. Project Management**

Project steering group**:** Annual meetings to provide strategic oversight and ensure successful delivery of my project aims. Members will include my patient partners, mentors, Shaun Rowark (NICE representative), Professor Abhishek (gout/trial expertise), Dr Chowla (GP), Dr Hudson (qualitative methodologist), Professor Norton (statistician), and Dr Ellis (policy advisor to Versus Arthritis).

Patient advisory group: Meetings at the beginning of each workstream, then annually, to oversee PPIE strategy, including co-developing educational materials, interpreting study results, and facilitating dissemination. Members will include patients, lay members, patient organisation representatives, and a PPIE coordinator from KCL.

Trial steering committee: 6-monthly meetings during Workstream 3 to oversee trial conduct/progress. Members will include an independent chair, statistician, GP, patient representative, and King's CTU representative.

Data monitoring and ethics committee: To safeguard trial participants, including monitoring of safety outcomes.

Mentors**:** Bi-monthly meetings with Professors Galloway, Roddy and Coates to oversee my training and development into an independent researcher.

Financial oversight with the KCL finance team to monitor my research budget.

**8. Ethics**

I have been granted approval by NHS England to conduct analyses of gout management using OpenSAFELY. I have conducted pilot analyses assessing the impact of the pandemic on gout care.[2] NHS England recently expanded OpenSAFELY's use beyond COVID-19-related projects, which covers the aims outlined in this proposal.[23]

I will submit ethics applications covering each workstream. I have experience of leading successful ethics applications for research projects in OpenSAFELY. Importantly, the privacy safeguards in OpenSAFELY are greater than for any other data platform in the UK.[20,53] No individual patient data are visible or transferred to researchers. Only summary-level data are outputted following strict disclosivity checks. OpenSAFELY adheres to the obligations of UK GDPR and the Data Protection Act 2018, and the data of patients with type 1 opt-outs will not be processed.

There is a precedent for conducting data-enabled, cluster-randomised trials in OpenSAFELY.[32] Following the principles outlined by Weijer *et al.,*[49,54] informed consent will be requested from a senior partner at eligible practices. The intervention will be implemented at practice-level, and all treatment decisions will remain the responsibility of the health professionals. No individual patient consent will be required.[49,54] Non-identifiable, individual-level data will be analysed according to the governance framework for OpenSAFELY.[20]

**9. Success criteria**

Key criteria I will use to determine success include:
- Year 1: Finalised analysis pipeline for monitoring guideline implementation, shared with collaborators at NICE;
- Year 2: Successful development of a multi-faceted intervention aligned to NICE recommendations;
- Years 3-5: Completion of a cluster-randomised trial in OpenSAFELY, with analysis code published via GitHub and adaptable for other interventions and diseases;
- High-impact publications and presentations for each workstream;
- Training course completion;
- Tenure via the King's Fellow-to-Faculty scheme
- Submission of an application for a follow-on Advanced Fellowship.

In each workstream, I have outlined potential risks identified with my collaborators, and how I will mitigate them. I will maintain a risk register, which will be reviewed with my project steering group.

## 7. Patient & Public Involvement

**Please describe how patients/service users, carers and the public have been involved in developing this proposal**

To generate ideas for my Fellowship, **I spoke with 10 people who had recently been treated for gout**, using semi-structured interviews to capture a diverse range of perspectives and priorities for improvement. The people I spoke with highlighted several barriers to better gout care. This included a lack of information on the role of preventative treatments; infrequent blood tests to monitor their urate levels; poor coordination of care; and a stigma that gout is a self-inflicted disease due to bad diet. People from under-served groups, particularly those with multiple long-term conditions, felt that their gout was viewed as unimportant, despite having a large impact on their quality of life. These are issues that I have encountered repeatedly for people with gout during my clinical career, despite numerous studies showing that the symptoms and complications of gout can be prevented effectively with guideline-recommended treatments such as allopurinol.

To develop my proposal further, I convened a **stakeholder meeting** and a **patient and public engagement event**, and collaborated with the leading patient organisation for people living with gout in the UK (the **UK Gout Society**). The feedback received was incredibly informative: whereas my previous research had focused on improving gout care in a hospital setting, it became clear that many of the difficulties patients continue to encounter are in the community (where most gout care is provided). Patients reiterated that a single, focused intervention would not address the many barriers to optimal gout care. Instead, a multi-faceted intervention, incorporating both patient-centred education and clinician-facing solutions (e.g. digital solutions that streamline care) would be essential. This has informed the focus and framework for my study plan.

**Two public partners** – **Joel and Erman** – who are experts by experience, have agreed to support me throughout my Fellowship. They have extensive experience of the issues my research seeks to address, and their input has been hugely beneficial in shaping this proposal, from study design to intervention ideas and dissemination plans.

**Please describe the ways in which patients/service users, carers and the public will be actively involved in the proposed research, including any training and support provided**

My **public partners** will be an integral part of my Fellowship. They will offer advice as members of my steering group, and will be invited to join our research team. The Centre for Rheumatic Diseases has an established process for arranging honorary research contracts and for supporting patient advisory groups. All patient involvement will be supported by appropriate reimbursement, budgeted for in this proposal.

With the support of the UK Gout Society, I will invite additional public partners to form an **advisory group**, with broad representation from under-served groups, lay members, patient organisations (including Dr Benjamin Ellis, policy advisor to Versus Arthritis), and a patient and public involvement (PPI) coordinator from King's. The group will oversee PPI strategy, including ethics approval documentation, co-development of educational materials and focus group topic guides, interpreting study results, dissemination planning, and a patient-led presentation at the British Society for Rheumatology (BSR) conference.

Patients will be invited to facilitate **focus groups** to explore areas of good care and areas for improvement. Training will be provided for patient facilitators, using the extensive PPI support available at King's. I will invite people with gout who participate in my study to a patient and public engagement event, to **share our findings, explore their perspectives, and identify areas for further improvement**.

I am a **trustee of the UK Gout Society**, who will help me reach people with gout throughout the UK via their membership, social media, and newsletters. In partnership with patients, the UK Gout Society and the BSR, I will develop educational videos for people with gout, covering the condition, its treatment, and self-management. These videos will form part of my study intervention, and will be publicised widely to ensure they benefit as many people living with gout as possible.

To disseminate my findings to people with gout and the public, I will use several approaches:

- Plain language summaries accompanying all publications, co-written with my patient advisory group;
- Social media summaries of outputs, including via podcasts and X/Twitter;
- Public engagement events, utilising NIHR, RCGP and BSR networks;
- Research open-days and science festivals;
- Newsletters (e.g. via UK Gout Society).

| **If it is considered not appropriate and meaningful to actively involve patients/service users, carers and the public in your proposed research, please justify why** |
| --- |
| N/A |

## 8. Training & Development and Research Support

**Proposed training and development programme**

Training in skills necessary to support the proposed research

My NIHR Doctoral Fellowship and Clinical Lectureship have provided me with a strong foundation in big data analysis, epidemiology and implementation science. An Advanced Fellowship would give me the opportunity to develop new skills that will be incredibly helpful for this research proposal and in my future career as an independent researcher in data-driven quality improvement:

- In Years 1 and 2, I will complete a part-time **MSc in Health Data Science at the London School of Hygiene and Tropical Medicine**. This course aligns closely with the skills I will be using in Workstreams 1 and 3 of my proposal. It will be a fantastic opportunity to advance my skills as a health data scientist, including modules dedicated to machine learning, predictive modelling and Bayesian analysis.
- In Year 2, I will complete the **Developing and Evaluating Complex Interventions course at the National Centre for Research Methods**, University of Southampton. This course is led by Dr Kathyrn Skivington, lead author of the MRC/NIHR Complex Intervention Development Framework, which I will be using to develop a complex intervention in Workstream 2 of my proposal. The skills I learn in this course will complement the training I received in Health Behaviour Change (University of Oxford) and Implementation Science (King's College London) during my Doctoral Fellowship.
- In Year 3, I will complete the **Design and Analysis of Cluster Randomised and Stepped Wedge Trials course at the London School of Hygiene and Tropical Medicine**. This course is led by Professor John Bradley, an expert in cluster randomised trials. It covers many of the skills I will be utilising in Workstream 3 of my proposal, including cluster randomisation procedures, participant- and cluster-level analysis of treatment effects, and ethical considerations in cluster randomised trials.

These courses will complement the **training and expertise available to me at King's College London**, including from experts in epidemiology (Professor James Galloway), data-enabled trial design (Professor Martin Gulliford), experimental medicine (Professor Andrew Cope), statistics (Professor Sam Norton), implementation science (Centre for Implementation Science), and PPIE (Dr Nicky Wilson).

Placements with other groups

During Year 1 of my Fellowship, I will complete a **3-month placement (1 day a week) with software developers at the Bennett Institute for Applied Data Science, University of Oxford**. This will give me the opportunity to learn one-on-one from the team of professional coders, database engineers and data visualisation experts behind the OpenSAFELY Trusted Research Environment (led by Professor Ben Goldacre), which I will be utilising in Workstreams 1 and 3.

Throughout my Fellowship, I will be working closely with the **Data Analytics and Implementation Support teams at NICE** to develop the infrastructure needed to monitor the real-world uptake of clinical guidelines. This will be an incredible learning and career development opportunity. It will enable me to influence the development and implementation of national guidelines across numerous disease areas.

Training to support my development as a future health research leader

I will apply to the **Emerging Research Leaders stream of NIHR's Future-Focused Leadership programme**, which is designed specifically for post-doctoral researchers taking up their first significant leadership and management roles. Through a mixture of residential workshops and training modules, I will be able to explore the management and leadership skills that are essential for a successful career in health research, including the ability to inspire other team members, build strong collaborations with external organisations, and the importance of strategic awareness.

To further my skills in PPIE, I will complete a series of workshops at the **National Coordinating Centre for Public Engagement**, including modules in community engagement and partnership working, in addition to the dedicated **PPIE training sessions at the NIHR BRC, University College London Hospitals**.

I will take full advantage of the training opportunities available to health researchers at King's College London, including training programmes in **Equality, Diversity and Inclusion, media engagement, and supervisory skills**.

Overseas research visit

In Year 4 of my Fellowship, I will complete a **two-week placement at the University of Tsukuba, Japan**, with Professor Masao Iwagami and his team. This will give me the opportunity to learn about data-driven quality improvement from clinical epidemiologists in an entirely different healthcare system, in a country that is renowned for its approach to continuous quality improvement. I will apply this learning to my research proposal, and share it with my collaborators at NICE, to build an understanding of how the UK can be an exemplar for data-driven quality improvement.

Conferences

In each year of my Fellowship, I will attend at least one national conference to present my research findings and learn from experts in the fields of **rheumatology** (British Society for Rheumatology Annual Conference), **primary care** (Royal College of General Practitioners Annual Conference) and **quality improvement** (NHS Providers Quality & Improvement Conference). I have also costed in attendance at two international conferences during my Fellowship, to give me the opportunity to engage with leading health researchers from Europe and the United States.

Planned clinical sessions

I am requesting 20% of my time to support two clinical sessions as a consultant rheumatologist. I will undertake a weekly clinic in inflammatory arthritis at King's College Hospital NHS Foundation Trust, and establish a monthly, multi-disciplinary, **crystal arthritis clinic** in collaboration with my colleagues at Guy's and St Thomas' Hospital NHS Foundation Trust. This clinic will provide best practice care for people with gout and other crystal arthropathies, which aligns closely with the aims of my Fellowship. There is a huge clinical need for this service, with very few dedicated clinics in the UK for people with severe gout (e.g. those with end-stage renal disease, severe tophaceous gout or recurrent hospitalisations). It will be fantastic opportunity to establish and lead such a service.

**RESEARCH SUPPORT / SUPERVISOR 1**

| **Name** | Professor James Galloway |
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| **Institution** | King's College London |
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| **Position** | Professor of Rheumatology |
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| **Description of support provided** |
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| James Galloway is a Professor of Rheumatology at King's College London and Honorary Consultant Rheumatologist at King's College Hospital NHS Foundation Trust. He is the lead methodologist for the National Early Inflammatory Arthritis Audit in England and Wales; methodology advisor to the British Society for Rheumatology; and Deputy Director of the King's College London Clinical Research Facility. Professor Galloway's expertise in epidemiology, quality improvement and clinical trials, make him ideally placed to support my research proposal and career development.<br><br>Professor Galloway has been a fantastic mentor to me throughout my clinical academic career. He was my primary supervisor during my NIHR Doctoral Fellowship, and is the academic supervisor for my NIHR Clinical Lectureship. He has been incredibly supportive of me in developing my own research niche in data-driven quality improvement, and in offering me his time and expertise as I developed this Advanced Fellowship proposal.<br><br>Professor Galloway is ideally placed to support my development into an independent health researcher. He has extensive experience as a supervisor and early career mentor, and he is Programme Director for Integrated Academic Training at King's College London. He has several senior management roles within higher education, and collaborates closely with national organisations including NHS England, CQC and HQIP, to whom my research is closely aligned. Importantly, Professor Galloway is familiar with the challenges of navigating clinical work as an NHS rheumatologist, alongside a successful academic career. |

**RESEARCH SUPPORT / SUPERVISOR 2**

| **Name** | Associate Professor Laura Coates |
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| **Institution** | University of Oxford |
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| **Position** | NIHR Clinician Scientist |
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| **Description of support provided** |
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| Laura Coates is an associate professor and previous NIHR Clinician Scientist at the University of Oxford, and honorary consultant rheumatologist at Oxford University Hospitals NHS Foundation Trust. She is based outside of my institution and will be less directly involved in the day-to-day running of my project, with a greater focus on my career development and progression.<br><br>Professor Coates is a fantastic role model for how to build a successful career as a health researcher. She has extensive experience of delivering successful grants, including a £1.8 million NIHR EME grant exploring precision medicine in psoriatic arthritis, and a £2.3 million NIHR HSDR grant evaluating whether patient-initiated care leads to improved outcomes in rheumatology. She is a co-lead on international consortia grants totalling over €30 million. She has a strong background in clinical trials and translational medicine, and has led several national and international clinical guidelines. She is a leading clinical academic in the management of psoriatic arthritis – an inflammatory joint condition that, like gout, is associated with multiple long-term health conditions (including metabolic syndrome) and adverse health outcomes if undertreated. During her NIHR Fellowship, she established an NHS service that delivers optimal ('treat-to-target') management for patients with psoriatic arthritis; something I hope to replicate for patients with gout during my Fellowship. |

**RESEARCH SUPPORT / SUPERVISOR 3**

| **Name** | Professor Edward Roddy |
|---|---|

| **Institution** | Keele University |
|---|---|

| **Position** | Professor of Rheumatology |
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| **Description of support provided** |
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| Edward Roddy is a Professor of Rheumatology at Keele University and Honorary Consultant Rheumatologist at the Haywood Hospital, Midlands Partnership University NHS Foundation Trust. He is Chair of the British Society for Rheumatology (BSR) Guidelines Steering Group.<br><br>Professor Roddy is a leading expert in the management of gout. He has led numerous, highly successful, research grants in this field, spanning epidemiology and clinical trials, including grants funded by the NIHR Health Technology Assessment, NIHR Research for Patient Benefit Programme, and NIHR School for Primary Care Research. He led the 2017 BSR Gout Guideline and was clinical topic advisor to the 2022 NICE Gout Guideline. He therefore has critical insight into the guideline I will be using to develop a data-driven approach to monitoring guideline uptake.<br><br>Professor Roddy's support and guidance was a key driver behind the success of my doctoral research. He has also provided external mentorship to other academic clinicians, resulting in successful NIHR doctoral and post-doctoral fellowship applications. His broad methodological expertise and experience in delivering successful research grants in the field of crystal arthritis make him an ideal mentor for me. His ongoing support will help to ensure the success of my fellowship proposal and my development into a leading clinical academic in the field of crystal arthritis. |

| Collaborations |
| --- |
| I have established a strong collaboration with the **National Institute for Health and Care Excellence (NICE)** to support my research proposal and career development. In Workstream 1, I will be collaborating closely with the Data Analytics team at NICE (led by Shaun Rowark) to develop a data-driven framework for monitoring the uptake and impact of national guidelines. In Workstreams 2 and 3, I will work with the Implementation Support team at NICE to develop interventions that bridge identified evidence-practice gaps, as well as helping to target implementation-support activities.<br><br>I am collaborating with the **Bennett Institute for Applied Data Science, University of Oxford** (led by Professor Ben Goldacre), who run the OpenSAFELY Trusted Research Environment I will be utilising in Workstreams 1 and 3 of my proposal. OpenSAFELY is the largest and most secure health data platform in the UK. As well as supporting the delivery of my research proposal, I will complete a 3-month training placement with the Bennett Institute, to learn from their team of software engineers and health data experts.<br><br>I am collaborating with **Clinical Effectiveness South East London** and **Dr Rachna Chowla** (General Practitioner and Joint Director of Clinical Strategy, King's Health Partners), who have extensive experience of implementing best practice care for long-term conditions in primary care. Their experience of combining education and digital solutions (e.g. decision-support tools via EMIS/Ardens) is very relevant to the intervention I will be developing and implementing in Workstreams 2 and 3. Additionally, I will be collaborating with **Dr Shalini Ahuja** (Lecturer in Implementation Science, King's College London), **Dr Joanna Hudson** (Lecturer and Health Psychologist, King's College London) and the **Centre for Implementation Science** (King's College London) to ensure my intervention and implementation strategies have a robust theoretical basis for enacting behaviour change.<br><br>I will be collaborating with **Professor Abhishek Abhishek** (Professor of Rheumatology, University of Nottingham) and **Professor Edward Roddy** (Professor of Rheumatology, Keele University) who have extensive experience of conducting trials to optimise gout management in primary care. I have engaged the support of **Professor Martin Gulliford** (Professor of Public Health and Epidemiology, King's College London), who has pioneered the use of data-enabled, cluster-randomised trials in primary care settings. Their combined expertise and support will be invaluable for the data-enabled, primary care-based trial I undertake in Workstream 3.<br><br>As well as my **patient partners**, I will be collaborating with the **UK Gout Society** throughout my Fellowship. The UK Gout Society are the leading patient organisation for people living with gout in the UK. Their input will help ensure that my research benefits as many people as possible with this highly treatable, yet poorly managed, condition. I have also engaged the support of **Dr Benjamin Ellis MBE, senior policy advisor to Versus Arthritis**, and consultant rheumatologist at Imperial College Healthcare NHS Trust. Dr Ellis's experience in driving improvements in healthcare for people with long-term conditions, particularly under-served groups, is highly relevant to the aims of my Fellowship.<br><br>During my Fellowship, I will complete a visiting research placement with **Professor Masao Iwagami at the University of Tsukuba, Japan.** This will enable me to build a collaboration with a team of clinical epidemiologists and quality improvement experts in a different healthcare system, and gain first-hand experience of how real-world data can drive improvements in patient care. |

**Host Organisation Support Statement**
The following statement of support must be completed by the head of your lead academic department. In order to contribute to this application, this individual must be invited to do so via the 'Participants and Signatories' section of the form.

| Statement of support |
| --- |
| We are delighted to support this exciting application, and to host Dr Mark Russell at the Centre for Rheumatic Diseases at King's College London.<br><br>**Why I support Dr Russell**<br><br>Mark is an excellent candidate for this NIHR Advanced Fellowship. Throughout his doctoral and post-doctoral research, Mark has demonstrated incredible drive and motivation to learn new skills and collaborate. He is conscientious, hard-working, and strives for the best, both in his academic work and for his patients. Mark's research outputs are testament to his dedication, having published 28 manuscripts in high-impact journals during his 3-year NIHR Doctoral Fellowship, including first-author publications in *The Lancet Rheumatology, The Lancet Regional Health,* and *Annals of Rheumatic Diseases*. He has been awarded national prizes, and was interviewed by the *BBC*, *The Times,* and the *Guardian* on the impact of his research for people with gout. |

Since joining our department in 2017, Mark has demonstrated outstanding initiative and independence for an early-career researcher. He has formed strong collaborations with organisations including *NICE*, the *Healthcare Quality Improvement Partnership*, and *NHS England*, and has developed an exciting research niche in data-driven quality improvement. He has taken full advantage of the opportunities at King's, including training in statistical methodology, implementation science, and leadership and management. He has taken on teaching and examination roles for our MBBS and Integrated Academic Training programmes, and he now plays a key role in using large-language models to develop MBBS examination questions.

Mark's recognition of the sub-optimal care offered to patients with gout has inspired him to use this condition to develop the methodology in this Fellowship proposal. Gout is a disease that impacts enormously on patients, for which there is a desperate need for better care, and for which Mark has a strong track-record in delivering successful research. Mark was awarded the prestigious *British Society for Rheumatology* Young Investigator Award for his demonstration of a large increase in hospitalisations for gout in England. He collaborated extensively with patients and other stakeholders to implement a pathway that optimised care for people hospitalised for gout. His research has directly benefitted patients, received national recognition, and has been adopted by other NHS hospitals.

Mark's Advanced Fellowship proposal has the potential to transform how we monitor the quality of care that is delivered to patients. He will deliver infrastructure that does not exist currently: the ability to evaluate care quality in near real-time. This is essential to improving patient outcomes and to organisations including *NICE*. Mark's unique skills in big data analysis and quality improvement methodology, combined with his clinical expertise, make him an ideal person to lead this work.

**Why King's is the ideal host institution for Mark**

The Centre for Rheumatic Diseases is a EULAR Centre of Research Excellence for rheumatology research. It has been a launchpad for success for multiple clinical academics in the field of rheumatology and health services research. Underlying the success of our department is cross-disciplinary leadership that bridges epidemiology, clinical trials and qualitative methodology, integrated within NHS clinical practice. We have extremely supportive team members, and an ethos of mentoring post-doctoral researchers through the next stage of their careers. We prioritise training and career development over the quantity of research outputs, and we believe in the importance of building a future generation of collaborative researchers who put patients at the centre of science.

Mark will benefit from our extensive experience of delivering high-quality health services research using routinely-collected healthcare data. He will receive training and methodological support from experts in clinical epidemiology (Professor James Galloway), statistical methodology (Professor Sam Norton), experimental medicine (Professor Andrew Cope), health psychology (Dr Joanna Hudson), and patient and public involvement (Dr Nicky Wilson). He will have a wealth of expertise and training opportunities available in complex intervention development via the Centre for Implementation Science at King's, as well the support of a leading expert in data-enabled clinical trials (Professor Martin Gulliford).

Mark will have access to a broad range of training in research methods, career development and mentorship opportunities at King's. Recognising that diversity of experience is essential at this stage of training, we actively encourage training placements with other institutions and industry. Our department's strong links with organisations including NHS England, the Healthcare Quality Improvement Partnership, and the British Society for Rheumatology, will provide Mark with the opportunity to become a leading collaborator in data-driven quality improvement.

We have a strong track record in delivering bridging support to clinical researchers on completing fellowships. We supported Mark in obtaining funding to bridge his transition from Doctoral Fellowship to Clinical Lecturer, and as a co-applicant on NIHR HSDR and MRC programme grants. We are committed to supporting Mark during his Advanced Fellowship, including through the provision of PhD students and research support. King's also has a Fellow-to-Faculty process which he would be eligible to pursue.

We are dedicated to ensuring equality of opportunity, recognising and celebrating our diversity and inclusion. We have been a member of the Athena SWAN Gender Equality Charter since 2007, and King's gained its most recent Silver institutional award in 2021. We hold a Bronze Race Equality Charter Mark Award, and we are members of the Stonewall Diversity Champions programme, Business Disability Forum and Working Families. We have a variety of staff networks that support our equality, diversity, and inclusion strategy, including NEST (parents and carers), Race Equality, Access King's (disability equality), Elevate (gender equality), and Proudly King's (LGBTQIA+ equality). We are holders of the European Commission's HR Excellence in Research Award, in recognition of our commitment to the principles of the Concordat to Support the Career Development for Researchers.

I strongly believe that Mark is an excellent candidate for this NIHR Advanced Fellowship. It will provide him with dedicated training and expertise in health data analysis and implementation science, and enable him to establish himself as an independent researcher in this field. We offer our unreserved support to Mark, and we are committed to ensuring that the infrastructure, training and research support are in place for him to succeed.

## 9. Detailed Budget

**The finance section should provide a breakdown of costs associated with undertaking the research as described in the proposal.**

**This section must be completed by the Applicant in conjunction with the Research Support Office or Finance Office at the host organisation.**

**1**

| | |
|---|---|
| **Staff type** | Lead Applicant |

| | |
|---|---|
| **Name** | Mark Russell |

| | |
|---|---|
| **Employing organisation** | King's College London |

| | |
|---|---|
| **Role on this research** | Fellow |

| | |
|---|---|
| **Grade** | Consultant |

| | |
|---|---|
| **Starting point on scale** | 2 |

| | |
|---|---|
| **Increment Date** | 01/08/2026 |

| **Justification of Costs** |
|---|
| As lead applicant, if successful, I would commence the Fellowship at consultant level. I have requested five years of salary (100% WTE), which includes basic salary, national insurance contributions, superannuation, London geographical allowance, and apprenticeship levy (under other allowances). |

**Details of Posts and Salaries Costs**

| | **Total (£)** |
|---|---|
| Salary (£) | 96230 |
| Geographical Weighting (£) | 2162 |
| Other allowances (£) | 492 |
| Superannuation (£) | 14267 |
| Nat. Insurance (£) | 12322 |
| **Current Annual Costs (£)** | **125473** |

**2**

| | |
|---|---|
| **Staff type** | Support Staff |

| | |
|---|---|
| **Name** | Post-doctoral research assistant |

| | |
|---|---|
| **Employing organisation** | King's College London |

| | |
|---|---|
| **Role on this research** | Research assistant |

| | |
|---|---|
| **Grade** | 6 |

| | |
|---|---|
| **Starting point on scale** | 31 |

| | |
|---|---|
| **Increment Date** | 01/08/2026 |

| **Justification of Costs** |
|---|
| I have requested salary for a post-doctoral research assistant with qualitative expertise at 50% FTE for years 2 to 5 of the award. In Workstream 2 (year 2), they will support the focus groups and intervention development work. In Workstream 3 (years 3 to 5), they will help implement the intervention in participating general practices, and conduct semi-structured interviews and thematic analysis in a nested qualitative study. I have requested the post to be at the early post-doctoral level to ensure they have sufficient experience of qualitative methodology and project management. |

**Details of Posts and Salaries Costs**

| | **Total (£)** |
|---|---|
| Salary (£) | 38205 |
| Geographical Weighting (£) | 5000 |
| Other allowances (£) | 216 |
| Superannuation (£) | 6265 |
| Nat. Insurance (£) | 4706 |
| **Current Annual Costs (£)** | **54392** |

---

**Staff Type:** (Lead Applicant)-Dr Mark Russell
**Type of cost:** HEI
**% Full-time on this research:** 100.0
**Total months on this research:** 60.0
**Justification of Cost:** As lead applicant, I would commence the Fellowship at consultant level. I have requested five years of salary (100% WTE), which includes basic salary, national insurance contributions, superannuation, London geographical allowance, and apprenticeship levy (under other allowances).

| | **Year 1** | **Year 2** | **Year 3** | **Year 4** | **Year 5** | **Total Cost (£)** |
|---|---|---|---|---|---|---|
| Basic Salary | 76984 | 80212 | 82557 | 84312 | 84312 | 408377 |
| National Insurance | 9858 | 10303 | 10627 | 10869 | 10869 | 52526 |
| Superannuation | 11414 | 11882 | 12222 | 12476 | 12476 | 60470 |
| Geographical weighting / Other allowances (if applicable) | 2123 | 2139 | 2151 | 2160 | 2160 | 10733 |

**Staff Type:** Other
**Other Type:** Support Staff

**Name:** Post-doctoral research assistant
**Type of cost:** HEI
**% Full-time on this research:** 50.0
**Total months on this research:** 48.0
**Justification of Cost:** I have requested salary for a post-doctoral research assistant with qualitative expertise at 50% FTE for years 2 to 5 of the award. In Workstream 2 (year 2), they will support the focus groups and intervention development work. In Workstream 3 (years 3 to 5), they will help implement the intervention in participating general practices, and conduct semi-structured interviews and thematic analysis in a nested qualitative study. I have requested the post to be at the early post-doctoral level to ensure they have sufficient experience of qualitative methodology and project management.

| | Year 1 | Year 2 | Year 3 | Year 4 | Year 5 | Total Cost (£) |
|---|---|---|---|---|---|---|
| Basic Salary | 0 | 15282 | 15701 | 16170 | 16653 | 63806 |
| National Insurance | 0 | 1882 | 1940 | 2005 | 2072 | 7899 |
| Superannuation | 0 | 2506 | 2566 | 2634 | 2705 | 10411 |
| Geographical weighting / Other allowances (if applicable) | 0 | 2086 | 2089 | 2091 | 2093 | 8359 |

| Totals | Applicant | Support Staff | Shared Staff | Total Cost (£) |
|---|---|---|---|---|
| Total HEI | 532106 | 90475 | 0 | 622581 |
| Total NHS | 0 | 0 | 0 | 0 |
| Total Commercial | 0 | 0 | 0 | 0 |
| Total Other | 0 | 0 | 0 | 0 |

**Travel, Subsistence and Conference Fees**

**Type of cost:** HEI
**Description:** Conference Fees
**Conference:** True
**Item Description:** Conferences
**Justification of Cost:** I have requested funds to attend a national conference in each year of my Fellowship, and two international conferences during my Fellowship. This will enable me to present my research findings and learn from experts in the fields of rheumatology (BSR, EULAR, ACR Conferences), primary care (RCGP Annual Conference) and quality improvement (NHS Providers Quality & Improvement Conference). The requested costs cover all related travel, accommodation and conference fees.

| | Year 1 | Year 2 | Year 3 | Year 4 | Year 5 | Total Cost (£) |
|---|---|---|---|---|---|---|
| | 800 | 800 | 2000 | 800 | 2000 | 6400 |

**Type of cost:** HEI
**Description:** Journey Costs
**Conference:** False
**Item Description:** Steering group meetings
**Justification of Cost:** I will convene a project steering group meeting in each year of my Fellowship. An in-person meeting will be convened in Year 1. Subsequent meetings will be held virtually via Microsoft Teams. I have requested travel costs for members from outside of London (Shaun Rowark; Professor Roddy; Professor Abhishek; Professor Coates) to attend the in-person meeting.

| | Year 1 | Year 2 | Year 3 | Year 4 | Year 5 | Total Cost (£) |
|---|---|---|---|---|---|---|
| | 320 | 0 | 0 | 0 | 0 | 320 |

**Type of cost:** HEI
**Description:** Journey Costs
**Conference:** False
**Item Description:** Overseas research visit

**Justification of Cost:** In year 4, I will undertake a two-week placement at the University of Tsukuba, Japan, with Professor Masao Iwagami and his team. This will give me the opportunity to learn about data-driven quality improvement from leading clinical epidemiologists in an entirely different healthcare system, which I will be able to share with my collaborators at NICE. Return flights from London to Tokyo: £750; accommodation for 14 nights (£90/night).

| | Year 1 | Year 2 | Year 3 | Year 4 | Year 5 | Total Cost (£) |
|---|---|---|---|---|---|---|
| | 0 | 0 | 0 | 1608 | 0 | 1608 |

| | Year 1 | Year 2 | Year 3 | Year 4 | Year 5 | Total Cost (£) |
|---|---|---|---|---|---|---|
| Travel, Subsistence and Conference Fees Total | 1120 | 800 | 2000 | 2408 | 2000 | 8328 |

**Equipment**

**Type of cost:** HEI
**VAT relief to be applied:** False
**Description:** Computer
**Price excluding VAT £:** 1000
**Expected life time:** 60
**Justification of Cost:** I have requested funds for a computer with sufficient processing power and memory to perform resource intensive calculations during analyses of complex health datasets in Workstreams 1 and 3 of my proposal.

| | Year 1 (exc VAT) | Year 2 (exc VAT) | Year 3 (exc VAT) | Year 4 (exc VAT) | Year 5 (exc VAT) | Total Cost (£) (exc VAT) |
|---|---|---|---|---|---|---|
| | 800 | 0 | 0 | 0 | 0 | 800 |

| | Year 1 | Year 2 | Year 3 | Year 4 | Year 5 | Total Cost (£) |
|---|---|---|---|---|---|---|
| Equipment Total | 800 | 0 | 0 | 0 | 0 | 800 |

**Summary**

| | Travel (excl Conferences) | Travel (Conferences) | Equipment |
|---|---|---|---|
| Total HEI | 1928 | 6400 | 800 |
| Total NHS | 0 | 0 | 0 |
| Total Commercial | 0 | 0 | 0 |
| Total Other | 0 | 0 | 0 |

**Consumables**
No costs have been added

**Patient and Public Involvement**

**Description:** Recording of educational videos
**Type of cost:** HEI
**Please describe fully:** Recording of educational videos
**Justification of Cost:** I have requested funding for professional filming of educational videos for people living with gout. I will create these videos in collaboration with the UK Gout Society. They will form part of the intervention package I develop in Workstream 2, and will be made publicly available through the UK Gout Society, British Society for Rheumatology, and Royal College of General Practitioners.

| | Year 1 | Year 2 | Year 3 | Year 4 | Year 5 | Total Cost (£) |
|---|---|---|---|---|---|---|
| | 0 | 2400 | 0 | 0 | 0 | 2400 |

**Description:** Patient advisory group meetings
**Type of cost:** HEI
**Please describe fully:** Patient advisory group meetings
**Justification of Cost:** I will convene a patient advisory group meeting at the beginning of each project workstream, then annually, to oversee PPIE strategy, including co-developing educational materials, focus group topic guides, interpretation of study results, and facilitating dissemination. Using the NIHR PPIE guidance, I have costed £50 per member per meeting to reimburse travel, food and time.

| | Year 1 | Year 2 | Year 3 | Year 4 | Year 5 | Total Cost (£) |
|---|---|---|---|---|---|---|
| | 480 | 240 | 240 | 240 | 240 | 1440 |

**Description:** Reimbursement for patient representatives
**Type of cost:** HEI
**Please describe fully:** Reimbursement for patient representatives
**Justification of Cost:** I have requested funding to reimburse the patient representatives in my project steering group and trial steering group meetings for the time required for pre-reading, evaluating documents, and contributing to discussions.

| | Year 1 | Year 2 | Year 3 | Year 4 | Year 5 | Total Cost (£) |
|---|---|---|---|---|---|---|
| | 80 | 80 | 160 | 160 | 160 | 640 |

**Description:** Training for patient facilitators for focus groups
**Type of cost:** HEI
**Please describe fully:** Training for patient facilitators for focus groups
**Justification of Cost:** The three focus groups I convene in Workstream 2 will each have a patient facilitator. I have requested funding to reimburse patients for the time required to facilitate these groups and undertake training.

| | Year 1 | Year 2 | Year 3 | Year 4 | Year 5 | Total Cost (£) |
|---|---|---|---|---|---|---|
| | 0 | 240 | 0 | 0 | 0 | 240 |

**Description:** Patient-led presentation at a national conference
**Type of cost:** HEI
**Please describe fully:** Patient-led presentation at a national conference
**Justification of Cost:** This funding will enable one of my patient partners to present our research findings from a patient perspective at the British Society for Rheumatology annual conference, covering registration fees, travel and accommodation

| | Year 1 | Year 2 | Year 3 | Year 4 | Year 5 | Total Cost (£) |
|---|---|---|---|---|---|---|
| | 0 | 0 | 0 | 0 | 320 | 320 |

**Description:** Patient and Public Engagement Event
**Type of cost:** HEI
**Please describe fully:** Patient and Public Engagement Event
**Justification of Cost:** In the final year of my Fellowship, I will organise a PPIE event at King's College London to disseminate my research findings and share best practice to patients and members of the public. I have costed in venue hire for this event.

| | Year 1 | Year 2 | Year 3 | Year 4 | Year 5 | Total Cost (£) |
|---|---|---|---|---|---|---|
| | 0 | 0 | 0 | 0 | 800 | 800 |

| | Year 1 | Year 2 | Year 3 | Year 4 | Year 5 | Total Cost (£) |
|---|---|---|---|---|---|---|
| Patient and Public Involvement Total | 560 | 2960 | 400 | 400 | 1520 | 5840 |

**Other Direct Costs**

**Description:** Randomisation service, King's College London Clinical Trials Unit
**Type of cost:** HEI
**Please describe fully:** Randomisation service, King's College London Clinical Trials Unit
**Justification of Cost:** This funding covers the costs of the randomisation service provided by the King's College

London Clinical Trials Unit, which I will require in Workstream 3 of my proposal (letter of support and associated costs attached).

| | Year 1 | Year 2 | Year 3 | Year 4 | Year 5 | Total Cost (£) |
|---|---|---|---|---|---|---|
| | 0 | 0 | 640 | 0 | 0 | 640 |

**Description:** OpenSAFELY Data administration costs
**Type of cost:** HEI
**Please describe fully:** OpenSAFELY Data administration costs
**Justification of Cost:** This covers the administrative costs for the ingestion of pseudonymised identifiers of general practices participating in my data-enabled cluster RCT and linkage with other data sources in OpenSAFELY.

| | Year 1 | Year 2 | Year 3 | Year 4 | Year 5 | Total Cost (£) |
|---|---|---|---|---|---|---|
| | 0 | 0 | 1200 | 0 | 0 | 1200 |

**Description:** Clinical Trials Unit Statistician Support
**Type of cost:** HEI
**Please describe fully:** Clinical Trials Unit Statistician Support
**Justification of Cost:** I have requested the support of a senior statistician (5% FTE) and junior statistician (25% FTE) to provide statistical input for the duration of the trial in my Fellowship (years 3 to 5), as advised by the King's College London Clinical Trials Unit (letter of support attached).

| | Year 1 | Year 2 | Year 3 | Year 4 | Year 5 | Total Cost (£) |
|---|---|---|---|---|---|---|
| | 0 | 0 | 17078 | 17434 | 17803 | 52315 |

| | Year 1 | Year 2 | Year 3 | Year 4 | Year 5 | Total Cost (£) |
|---|---|---|---|---|---|---|
| Other Direct Costs Total | 0 | 0 | 18918 | 17434 | 17803 | 54155 |

**Training And Development**

**Type of cost:** HEI
**Description:** Short Courses
**Item Description:** Health Data Science MSc, London School of Hygeine and Tropical Medicine
**Justification of Cost:** In Years 1 and 2, I will complete a part-time MSc in Health Data Science at the London School of Hygiene and Tropical Medicine. This course aligns closely with the methodology I will be utilising in Workstreams 1 and 3. It will be a fantastic opportunity to advance my skills as a health data scientist, with modules dedicated to Bayesian analysis, predictive modelling and machine learning. This will be incredibly helpful for my career in big data analysis.

| | Year 1 | Year 2 | Year 3 | Year 4 | Year 5 | Total Cost (£) |
|---|---|---|---|---|---|---|
| | 6470 | 6470 | 0 | 0 | 0 | 12940 |

**Type of cost:** HEI
**Description:** Short Courses
**Item Description:** Developing and Evaluating Complex Interventions, University of Southampton
**Justification of Cost:** In Year 2, I will complete the Developing and Evaluating Complex Interventions course at the National Centre for Research Methods (University of Southampton). This course is led by Dr Kathyrn Skivington, lead author of the MRC/NIHR Complex Intervention Development Framework, which I will be using to develop a complex intervention in Workstream 2

| | Year 1 | Year 2 | Year 3 | Year 4 | Year 5 | Total Cost (£) |
|---|---|---|---|---|---|---|
| | 0 | 188 | 0 | 0 | 0 | 188 |

**Type of cost:** HEI
**Description:** Short Courses
**Item Description:** Design and Analysis of Cluster Randomised and Stepped Wedge Trials, London School of Hygiene and Tropical Medicine
**Justification of Cost:** In Year 3, I will complete the Design and Analysis of Cluster Randomised and Stepped Wedge Trials course at the London School of Hygiene and Tropical Medicine. This course is led by Professor John Bradley, an expert in cluster randomised trials. It covers many of the skills I will be utilising in Workstream 3 of my

proposal, including cluster randomisation procedures, participant-level and cluster-level analysis of treatment effects, and ethical considerations of cluster RCTs.

| | Year 1 | Year 2 | Year 3 | Year 4 | Year 5 | Total Cost (£) |
|---|---|---|---|---|---|---|
| | 0 | 0 | 1528 | 0 | 0 | 1528 |

**Type of cost:** HEI
**Description:** Short Courses
**Item Description:** Training placement with software developers at Bennett Institute, University of Oxford
**Justification of Cost:** In Year 1, I will complete a 3-month training placement (1 day a week) with software developers from the Bennett Institute, University of Oxford. This will give me the opportunity to learn one-on-one from the team of professional coders, database engineers and data visualisation experts at the OpenSAFELY Trusted Research Environment, which I will be utilising in Workstreams 1 and 3.

| | Year 1 | Year 2 | Year 3 | Year 4 | Year 5 | Total Cost (£) |
|---|---|---|---|---|---|---|
| | 5000 | 0 | 0 | 0 | 0 | 5000 |

| | Year 1 | Year 2 | Year 3 | Year 4 | Year 5 | Total Cost (£) |
|---|---|---|---|---|---|---|
| Training And Development Total | 11470 | 6658 | 1528 | 0 | 0 | 19656 |

| | Year 1 | Year 2 | Year 3 | Year 4 | Year 5 | Total Cost (£) |
|---|---|---|---|---|---|---|
| Total Direct Costs | 114329 | 136710 | 152699 | 152959 | 154663 | 711360 |

**Indirect costs**

**HEI Indirect costs**

**Type of cost:** Estates Charges
**Name of Higher Education Institute:** King's College London
**Justification of Cost:** Estate costs, including infrastructure technician costs, have been calculated using the TRAC methodology

| | Year 1 | Year 2 | Year 3 | Year 4 | Year 5 | Total Cost (£) |
|---|---|---|---|---|---|---|
| | 21401 | 32101 | 35224 | 35224 | 35224 | 159174 |

**Type of cost:** Other Indirect Costs
**Name of Higher Education Institute:** King's College London
**Justification of Cost:** Indirect costs have been calculated using the TRAC methodology

| | Year 1 | Year 2 | Year 3 | Year 4 | Year 5 | Total Cost (£) |
|---|---|---|---|---|---|---|
| | 55796 | 83694 | 100433 | 100433 | 100433 | 440789 |

| | Year 1 | Year 2 | Year 3 | Year 4 | Year 5 | Total Cost (£) |
|---|---|---|---|---|---|---|
| HEI Indirect costs Total | 77197 | 115795 | 135657 | 135657 | 135657 | 599963 |

**Commercial Indirect Costs**
No costs have been added

**Other Partner Organisation Indirect costs**
No costs have been added

| | Year 1 | Year 2 | Year 3 | Year 4 | Year 5 | Total Cost (£) |
|---|---|---|---|---|---|---|
| Total Indirect Costs | 77197 | 115795 | 135657 | 135657 | 135657 | 599963 |

**Research Costs Requested From Funder**

| | Direct costs | Indirect costs/Overheads | Amount requested |
|---|---|---|---|
| Total Higher Education Institution Costs (80%) | 685304 | 599963 | 1285267 |
| Total Higher Education Institution Costs (100%)s | 26056 | 0 | 26056 |
| Total Commercial Costs | 0 | 0 | 0 |
| Total Other Parther Organisation Costs | 0 | 0 | 0 |
| Total NHS Research Costs | 0 | 0 | 0 |
| **Total Research Costs Requested From Funder** | 711360 | 599963 | 1311323 |

**Summary**

| | Year 1 | Year 2 | Year 3 | Year 4 | Year 5 | Total Cost (£) |
|---|---|---|---|---|---|---|
| **Salaries** | **100379** | **126292** | **129853** | **132717** | **133340** | **622581** |
| **Travel-subsistence-and-conference-fees** | **1120** | **800** | **2000** | **2408** | **2000** | **8328** |
| **Equipment** | **800** | **0** | **0** | **0** | **0** | **800** |
| **Consumables** | **0** | **0** | **0** | **0** | **0** | **0** |
| **Patient-and-Public-Involvement** | **560** | **2960** | **400** | **400** | **1520** | **5840** |
| **Other-Direct-Costs** | **0** | **0** | **18918** | **17434** | **17803** | **54155** |
| **Training And Development** | **11470** | **6658** | **1528** | **0** | **0** | **19656** |
| **HEI-Indirect-costs** | **77197** | **115795** | **135657** | **135657** | **135657** | **599963** |
| **Commercial-Indirect-Costs** | **0** | **0** | **0** | **0** | **0** | **0** |
| **Other-Partner-Organisation-Indirect-costs** | **0** | **0** | **0** | **0** | **0** | **0** |
| **Total** | **191526** | **252505** | **288356** | **288616** | **290320** | **1311323** |

**NHS Support Costs**

| Cost Item: | Year 1 | Year 2 | Year 3 | Year 4 | Year 5 | Total Cost (£) |
|---|---|---|---|---|---|---|
| NHS Service Support Cost | 0 | 756 | 447 | 0 | 176 | 1379.0 |

| **Justification:** |
|---|
| **NHS Service Support Cost:** I have provided the NHS Support Costings for the following activities in my research proposal: focus groups (Workstream 2); training in the study intervention for general practice staff (Workstream 3); qualitative interviews (Workstream 3); and obtaining informed consent. These costs were calculated with an AcoRD specialist, and in discussion with the Primary Care lead at the NIHR RDN South London. |

| | Year 1 | Year 2 | Year 3 | Year 4 | Year 5 | Total Cost (£) |
|---|---|---|---|---|---|---|
| NHS Support Costs Total | 0 | 756 | 447 | 0 | 176 | 1379 |

| | |
|---|---|
| **Have you discussed and agreed these costs with the Lead Network?** | Yes |

**NHS Treatment Costs**
No costs have been added

| **Have you discussed and agreed the NHS costs with the Lead Trust?** | Yes |
|---|---|

| **Is the patient/service user care being provided different from the usual standard treatment for the condition?** |
|---|
| No |

**Usual Treatment Costs**
No costs have been added**Excess Treatment costs/(savings)**

| | Cost per patient(£) | Year 1 | Year 2 | Year 3 | Year 4 | Year 5 | Total Cost (£) |
|---|---|---|---|---|---|---|---|
| Total NHS Excess Treatment costs/(savings) | 0 | 0 | 0 | 0 | 0 | 0 | 0 |

**NHS Support and Excess Treatment Costs**

| | Total (£) |
|---|---|
| NHS Support Costs requested from networks | 1379 |
| NHS Excess Treatment costs/savings | 0 |
| TOTAL NHS SUPPORT AND TREATMENT COSTS | 1379 |

**Total Funding Required (including non-research costs)**

| | Total (£) |
|---|---|
| Total Research Costs Requested from Funder | 1311323 |
| Total NHS (Support and Excess Treatment) Costs | 1379 |
| Total Funding Required | 1312702 |

## 10. Uploads

(References - References.pdf) is included as an appendix within this file.

(Figures / Tables - NIHR Advanced Fellowship Figure.pdf) is included as an appendix within this file.

(Research Timetable - Gantt Chart NIHR Advanced Fellowship.pdf) is included as an appendix within this file.

(CTU Letter of Support - Letters of Support.pdf) is included as an appendix within this file.

**SOECAT FORM 1**

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## 11. Participants and Signatories

***Devolved Administrations***

no Devolved Administrations Participant has been added

***Post-Doctoral Primary Supervisor***

***Research Support***

Research Support Participant

| | |
|---|---|
| **Title** | Dr |
| **Forename(s)** | Laura |
| **Surname** | Coates |
| **Position** | Senior Lecturer |

Research Support Participant

| | |
|---|---|
| **Title** | Professor |
| **Forename(s)** | James |
| **Surname** | Galloway |
| **Position** | Professor / Honorary Consultant |

Research Support Participant

| | |
|---|---|
| **Title** | Dr |
| **Forename(s)** | Edward |
| **Surname** | Roddy |
| **Position** | Reader in Rheumatology/Honorary Consultant Rheumatologist |

***Host Organisation - Heads of Department***

Head of Department or Senior Manager

| | |
|---|---|
| **Title** | Dr |
| **Forename(s)** | Tanya |
| **Surname** | Shaw |
| **Position** | Senior Lecturer |

***Administrative Authority or Finance Officer***

Administrative Authority or FO Participant

| | |
|---|---|
| **Title** | Dr |
| **Forename(s)** | Antoni |
| **Surname** | Wojcik |
| **Position** | |

## 12. Acknowledgement, Review and Submit

**Conflict checks:**
Please declare any conflicts or potential conflicts of interest that you may have in undertaking this research, including any relevant, non-personal & commercial interest that could be perceived as a conflict of interest.

**Agreement to Terms and Conditions:**

I confirm that I have secured all necessary licences and ethical approvals in relation to the research and will abide by the terms of those licences and approvals in carrying out the research.

I confirm that I have read the NIHR Carbon Reduction Guidelines and, where possible, taken steps to reduce the carbon emissions generated by this research.

I confirm that all material within this application, with the exception of sections that other participants are required to complete, is my own work and I have taken steps to ensure that no parts of the application have been inappropriately plagiarised.

I undertake to conduct the research according to such guidelines for good research practice as the Host Organisation may from time to time laydown according to the NIHR Research Governance Framework.

If you are applying for an NIHR award which comes with partnership funding from another organisation (e.g. a charity or industry partner) your information (including your NIHR Application Form and subsequent progress/annual reports) may be shared with the partner organisation as part of the review process and ongoing monitoring of the award.

In completing this, I declare that, to the best of my knowledge, the information provided in this application, is true, accurate and complete. I have read the Guidance Notes and agree to accept the process by which an application is assessed and agree to abide by the conditions under which an award may be granted.

☒Confirmed

## Appendices

1. References.pdf
2. NIHR Advanced Fellowship Figure.pdf
3. Gantt Chart NIHR Advanced Fellowship.pdf
4. Letters of Support.pdf

# Appendix #1: References.pdf

**References**


1. Russell MD, Galloway JB, Andrews CD, et al. Incidence and management of inflammatory arthritis in England before and during the COVID-19 pandemic: a population-level cohort study using OpenSAFELY. *The Lancet Rheumatology* 2022; **4**(12): e853-e63.
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3. NICE. Measuring the use of NICE guidance. 2024. https://www.nice.org.uk/about/what-we-do/into-practice/measuring-the-uptake-of-nice-guidance
4. Yates M, Bechman K, Dennison EM, et al. Data quality predicts care quality: findings from a national clinical audit. *Arthritis Res Ther* 2020; **22**(1): 87.
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6. National Institute for Health and Care Research. NICE strategy 2021 to 2026. https://www.nice.org.uk/Media/Default/Get-involved/Meetings-In-Public/Public-board-meetings/Mar-24-pbm-NICE-strategy-2021-2026.pdf.
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# Appendix #2: NIHR Advanced Fellowship Figure.pdf

# The GuideALIGN Study: A data-driven approach to monitoring the implementation of NICE guidance

**The problem:** We cannot easily monitor whether guidelines are being followed in real-world practice or how this impacts on health outcomes

**Aim:** To develop a data-driven approach to monitoring real-world uptake of NICE guidance, highlighting healthcare inequality and the impact on health outcomes

## **Workstream 1.** Monitoring the implementation of NICE guidance

**OpenSAFELY** — **Anonymised health data for 58 million people in England**

**Develop this methodology using the NICE Gout Guideline then adapt it for other guidelines**

1. **Systematically monitor the uptake of NICE recommendations**
2. **Highlight variation by region and for under-served groups**
3. **Evaluate the impact of care quality on health outcomes**
4. **Use dashboards to feedback performance against NICE standards**

## **Workstream 2.** Develop an intervention to improve care for people with gout

**Intervention Development**

- Stakeholder input
- Person-based approach
- Focus groups
- Logic models
- Implementation scientists
- Think-aloud interviews

→

**Intervention Components**

- Electronic health record solutions (ardens HEALTHCARE INFORMATICS)
- Dashboards (OpenSAFELY)
- Educational content (UK Gout Society)

## **Workstream 3.** Data-enabled, cluster-randomised trial in OpenSAFELY

General practices (n=38) → Randomised 1:1 → Intervention from WS2 / Usual care

**All outcomes captured automatically in OpenSAFELY**

**Primary outcome:** Initiation of urate-lowering therapy, as per NICE

**Secondary outcomes:**
- Urate target attainment
- Medication adherence
- Hospitalisations
- Consultation rate
- Comorbidities

Granular implementation outcomes obtained via a nested qualitative study

## Outputs, impact and dissemination

Reduced pain, disability & complications for patients

**NICE** — Mechanism for monitoring real-world guideline uptake

Care Quality Commission — Ability to detect unwarranted variation in care quality

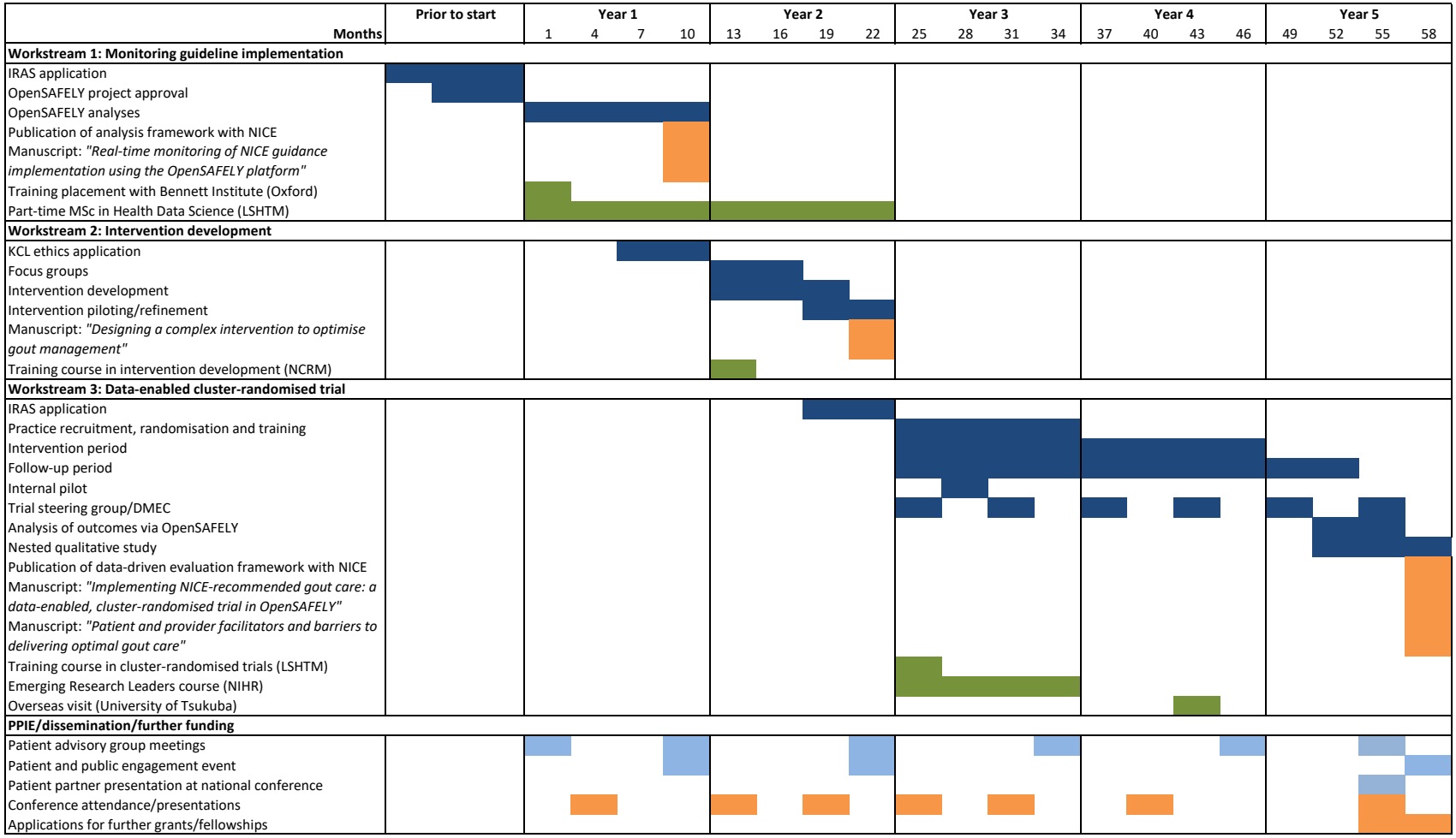
GitHub — Methodology shared openly to support efficient re-use

## Training and development

- MSc in Health Data Science
- Complex intervention development course
- Cluster-randomised trial course
- Training placement at OpenSAFELY
- Collaboration with NICE
- Emerging Research Leaders programme
- Overseas visit to Tsukuba University
- PPI and EDI workshops
- Conferences (BSR, EULAR, ACR, RCGP, QI)

King's College London · NICE · Bennett Institute for Applied Data Science · Clinical Effectiveness South East London · UK Gout Society

# Appendix #3: Gantt Chart NIHR Advanced Fellowship.pdf


Prior to start
Year 1
Year 2
Year 3
Year 4
Year 5
Months
1 4 7 10 13 16 19 22 25 28 31 34 37 40 43 46 49 52 55 58
Workstream 1: Monitoring guideline implementation
IRAS application
OpenSAFELY project approval
OpenSAFELY analyses
Publication of analysis framework with NICE
Manuscript: "Real-time monitoring of NICE guidance implementation using the OpenSAFELY platform"
Training placement with Bennett Institute (Oxford)
Part-time MSc in Health Data Science (LSHTM)
Workstream 2: Intervention development
KCL ethics application
Focus groups
Intervention development
Intervention piloting/refinement
Manuscript: "Designing a complex intervention to optimise gout management"
Training course in intervention development (NCRM)
Workstream 3: Data-enabled cluster-randomised trial
IRAS application
Practice recruitment, randomisation and training
Intervention period
Follow-up period
Internal pilot
Trial steering group/DMEC
Analysis of outcomes via OpenSAFELY
Nested qualitative study
Publication of data-driven evaluation framework with NICE
Manuscript: "Implementing NICE-recommended gout care: a data-enabled, cluster-randomised trial in OpenSAFELY"
Manuscript: "Patient and provider facilitators and barriers to delivering optimal gout care"
Training course in cluster-randomised trials (LSHTM)
Emerging Research Leaders course (NIHR)
Overseas visit (University of Tsukuba)
PPIE/dissemination/further funding
Patient advisory group meetings
Patient and public engagement event
Patient partner presentation at national conference
Conference attendance/presentations
Applications for further grants/fellowships

# Appendix #4: Letters of Support.pdf

# King's Clinical Trials Unit (KCTU)

NIHR | Research Support Service

*This unit receives National Institute for Health Research RSS Funding.*

**Professor Richard Emsley, KCTU Director**
**Caroline Murphy, KCTU Operational Director**

*King's Clinical Trials Unit • PO Box 64 • 16 De Crespigny Park • Denmark Hill • London SE5 8AF*
*TEL: + 44 (0)20 7848 0532 • FAX: +44 (0)20 7848 5229 • www.ctu.co.uk*

DATE 18-06-24

Dear Dr. Russell,

This is to confirm that the King's Clinical Trials Unit, King's College London (UKCRC registration number 053) have considered your request for support for;

**Study title:** A data-driven approach to monitoring the implementation of NICE guidance
**Funding Stream:** Post Doctoral Fellowship
**Application Stage:** one stage

The CTU will provide:

| Service/Collaboration | Service cost or collaborator name | Type of service or FTE for collaborator |
|---|---|---|
| Randomisation service | £800 excl. VAT | Service type Silver |
| Statistics supervised by | Sam Norton<br>Junior Statistician | 5%FTE Trial Duration<br>25%FTE Trial Duration |

We will be delighted to support you in this bid.

*Please note prices above are not inclusive of VAT.*

Yours sincerely

Caroline Murphy

Operational Director, King's Clinical Trials Unit

NICE National Institute for Health and Care Excellence

Level 1A
City Tower
Manchester
M1 4BT
United Kingdom

+44 (0)300 323 0140

NIHR
21 Queen Street
Leeds
LS1 2TW

13 June 2024

Dear NIHR Advanced Fellowship Committee,

**Letter of support for "A data-driven approach to monitoring the implementation of NICE guidance" from Dr Mark Russell.**

On behalf of the National Institute for Health and Care Excellence (NICE) I am pleased to support the proposed research proposal on using real-world data to track the implementation and impact of NICE guidance, led by Dr Mark Russell at King's College London. We believe in the use of real-world data to transform healthcare and it has been identified as a priority area for development within NICE.

We believe that the proposal, with its aims to track the implementation of our guidance using real-world data, can help us to understand where there is variation in care, which will enable us to direct implementation activities as well as potentially update our guidance if feedback suggests there are challenges in its implementation. This is aligned with our vision to help practitioners and commissioners get the best care to patients fast while ensuring value for the taxpayer. The use of real-world data in understanding the implementation and impact of our products is vital to NICE's strategy.

Over the past 21 years NICE has established itself as an international leader in technology evaluation, guideline development and evidence synthesis. Our work today spans three ecosystems (life sciences, guidelines, and information) that involve close working with partners to ensure patients have access to the latest technologies, advice and guidance.

We do this by:

1. producing useful and usable guidance for health and care practitioners
2. providing rigorous, independent assessment of complex evidence for new health technologies

3. developing recommendations that focus on what matters most and drive innovation into the hands of health and care practitioners

4. encouraging the uptake of best practice to improve outcomes for everyone.

We look forward to partnering with Dr Mark Russell on this where required and we are happy to attend any events, working groups or meetings relating to this work.

Therefore, we are delighted to commend this bid to NIHR, as we see this proposal as crucial to understanding the implementation and impact of our work.

Yours faithfully,

Shaun Rowark, Associate Director for Data Access and Analysis

UK Gout Society

PO Box 90, Hindhead GU27 9FW  Email: info@ukgoutsociety.org or visit us at: www.ukgoutsociety.org

17 June 2024

NIHR
21 Queen Street
Leeds
LS1 2TW

**Re: Letter of support: Dr Mark Russell, NIHR Advanced Fellowship**

Dear Sir/Madam,

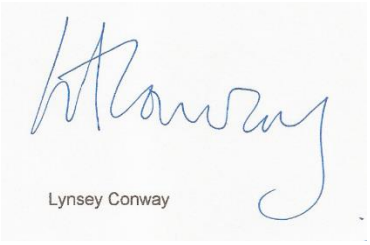
We are writing in support of Dr Russell's Fellowship proposal, entitled: "A data-driven approach to monitoring the implementation of NICE guidance".

The UK Gout Society is a leading charity for people living with gout. Our goal is to increase awareness and improve care for this painful and disabling, but very treatable, disorder. The UK Gout Society is a member of ARMA (Arthritis and Musculoskeletal Alliance) – the umbrella body providing a collective voice for the arthritis and musculoskeletal community in the UK. www.ukgoutsociety.org

Gout is the most common inflammatory arthritis, affecting 1 in 40 people in the UK. There are highly effective treatments that can prevent symptoms and complications in people with gout. However, Dr Russell's previous research has highlighted the urgent need for strategies to improve the uptake of these treatments and deliver better care for patients.

We look forward to supporting Dr Russell with this proposal.

Yours sincerely,




Signed on behalf of the UK Gout Society Board of Trustees
By Lynsey Conway
**UK Gout Society Secretariat**
**info@ukgoutsociety.org**

* The UK Gout Society is not responsible for the content of external Internet sites

UK Registered Charity Number 1093748

BENNETT INSTITUTE FOR APPLIED DATA SCIENCE

NUFFIELD DEPARTMENT OF PRIMARY CARE HEALTH SCIENCES
Medical Sciences Division

UNIVERSITY OF OXFORD

**Re: Letter of support: Dr Mark Russell, NIHR Advanced Fellowship**

In my capacity as Director of Research, Informatics and NHS Service Analytics at the Bennett Institute for Applied Data Science, University of Oxford, I am delighted to support Dr Russell's Fellowship proposal, entitled: "A data-driven approach to monitoring the implementation of NICE guidance".

At the Bennett Institute, we have pioneered the development of OpenSAFELY - a highly secure, transparent, software platform that facilitates the analysis of electronic health records for 58 million people. The research generated using OpenSAFELY has been fundamental to understanding of disease risk, vaccine uptake, and changes in patient care during the pandemic. We have worked closely with Dr Russell to evaluate the impact of the pandemic on care quality for people with chronic rheumatological diseases, including rheumatoid arthritis.

The capability to analyse health data for more than 99% of England's population in near real-time, makes OpenSAFELY ideally suited to monitoring the implementation of NICE guidance.

We look forward to collaborating with Dr Russell in this Fellowship proposal.

Kind regards,

Brian

**Specialist Pharmacist Adviser & Medicines Data Clinical Lead**
NHS England
brian.mackenna@nhs.net

**Director of NHS Service Analytics**
Bennett Institute for Applied Data Science
Nuffield Dept of Primary Care Health Sciences
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Clinical Effectiveness
South East London

National Institute for Health and Care Research

21 Queen Street

Leeds

LS1 2TW

12th June 2024

**Re: Support for Fellowship Proposal**

To whom it may concern

**Dr Mark Russell**

**NIHR Advanced Fellowship**

**A data-driven approach to monitoring the implementation of NICE guidance**

On behalf of the Clinical Effectiveness Southeast London (CESEL) Team please accept this letter in support of Dr Russell's Fellowship proposal.

CESEL is a quality improvement programme that supports general practice teams and primary care networks across southeast London with an aligned set of evidence-based tools, using local data and shared learning to support improvements in primary care across a range of long term conditions. Our aims align closely with those of Dr Russell in improving patient outcomes in common conditions and we wholeheartedly support his proposal.

Kind regards

Siân Howell

MB BS MRCGP MSc(Public Health)

Clinical Lead for CESEL

KING'S
HEALTH
PARTNERS

King's Health Partners
Counting House
Great Maze Pond
London Bridge
SE1 9RT

11th June 2024

National Institute for Health and Care Research
21 Queen Street
Leeds
LS1 2TW

**Re: Letter of support – Dr Mark Russell, NIHR Advanced Fellowship**

Dear Sir/Madam,

We would be delighted to support Dr Russell's Fellowship proposal, entitled: "A data-driven approach to monitoring the implementation of NICE guidance".

King's Health Partners via King's College London, host 'Clinical Effectiveness South East London' (CESEL) - a peer to peer GP evidence-based, data-driven quality improvement programme. Their goal is to improve health outcomes and reduce inequalities in care for our local population – which aligns very closely with this fellowship. In addition, King's Health Partners three strategic areas of focus include 'Population Health' and 'Health Data Sciences' where work aligns closely with the aims of Dr Russell's Fellowship proposal, and we look forward to collaborating with Dr Russell on this project.

Yours sincerely,

Dr Rachna Chowla
Joint Director of Clinical Strategy
King's Health Partners

University of Nottingham
UK | CHINA | MALAYSIA

12th June 2024

Fellowship Committee,
NIHR,
London, UK.

**Subject: Dr Mark Russell's application for NIHR Advanced Fellowship**

Dear Madam/Sir,

I am writing to you in support of Dr Mark Russell's application for an NIHR Advanced Fellowship.

The overall purpose of his project titled "A data-driven approach to monitoring the implementation of NICE guidance" is to systematically evaluate the uptake of guideline recommendations using real-world data, and improve management of people with gout. This research proposal aims to answer key questions in the field of gout and the findings have implications for GPs and rheumatologists.

I am a Professor of Rheumatology leading the Academic Rheumatology team at The University of Nottingham, UK. My research interests focuses on gout, pragmatic clinical trials and real world evaluation of evidence. During the proposed project, I will provide clinical and epidemiological expertise to the multidisciplinary research team. I will also provide Dr Russell with links to the national and international crystal arthritis research community via my leadership role in the British Society of Rheumatology crystal arthritis special interest group, European Crystal Network (Europe) and Gout-Crystal Arthritis Network (USA).

Yours sincerely,

Prof Abhishek, MBBS, MD, FRCP, PhD
Professor of Rheumatology
The University of Nottingham